

ARC–MOF: A Diverse Database of Metal–Organic Frameworks with DFT-Derived Partial Atomic Charges and Descriptors for Machine Learning

Jake Burner, Jun Luo, Andrew White, Adam Mirmiran, Ohmin Kwon, Peter G. Boyd, Stephen Maley, Marco Gibaldi, Scott Simrod, Victoria Ogden, and Tom K. Woo*



Cite This: *Chem. Mater.* 2023, 35, 900–916



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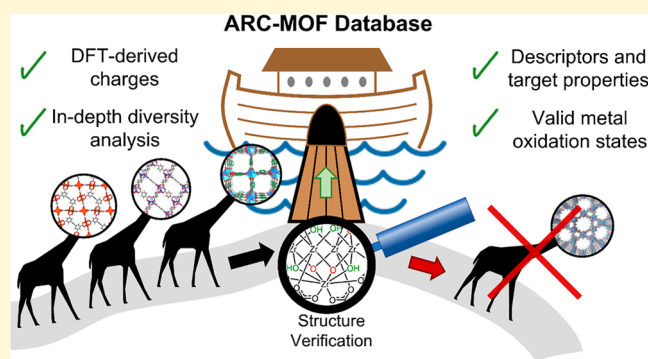
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ABSTRACT: Metal–organic frameworks (MOFs) are a class of crystalline materials composed of metal nodes or clusters connected via semi-rigid organic linkers. Owing to their high-surface area, porosity, and tunability, MOFs have received significant attention for numerous applications such as gas separation and storage. Atomistic simulations and data-driven methods [e.g., machine learning (ML)] have been successfully employed to screen large databases and successfully develop new experimentally synthesized and validated MOFs for CO₂ capture. To enable data-driven materials discovery for any application, the first (and arguably most crucial) step is database curation. This work introduces the ab initio REPEAT charge MOF (ARC–MOF) database. This is a database of ~280,000 MOFs which have been either experimentally characterized or computationally generated, spanning all publicly available MOF databases. A key feature of ARC–MOF is that it contains density functional theory-derived electrostatic potential fitted partial atomic charges for each MOF. Additionally, ARC–MOF contains pre-computed descriptors for out-of-the-box ML applications. An in-depth analysis of the diversity of ARC–MOF with respect to the currently mapped design space of MOFs was performed—a critical, yet commonly overlooked aspect of previously reported MOF databases. Using this analysis, balanced subsets from ARC–MOF for various ML purposes have been identified, with a case study of the effect of training set on the ML performance. Other chemical and geometric diversity analyses are presented, with an analysis on the effect of the charge-assignment method on atomistic simulation of the gas uptake in MOFs.



INTRODUCTION

Metal–organic frameworks (MOFs) are a class of crystalline materials composed of organic and inorganic structural building units (SBUs) and are often synthesized via self-assembly. A hallmark of MOFs is their nanoporosity resulting in record specific surface areas, with a theoretical upper limit of at least 14,600 m²/g.¹ Another key feature of MOFs is their functional tunability, arising from the fact that there is a seemingly infinite number of possible combinations of organic and inorganic SBUs, resulting in a virtually boundless design space. In addition to the chemical diversity of MOFs arising from the identity of the SBUs, the SBUs can be arranged in different net topologies allowing for significant geometric diversity. Thousands of distinct network topologies have been identified in experimentally characterized materials.² Due to these exceptional properties (large-surface areas, high porosity, and tunability), MOFs have attracted significant interest for a wide variety of applications, ranging from catalysis to drug delivery.^{3–6} With the urgent and serious concerns of climate change, one high-profile application of MOFs has been in gas

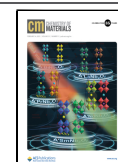
separation and storage,^{7–11} where MOFs are now being commercialized for industrial-scale CO₂ capture from combustion flue gases.^{12,13}

Rational design of materials for a specific application is more desirable than trial-and-error development, which can be a long and expensive process. To this end, various atomistic simulation methods such as density functional theory (DFT) and molecular dynamics (MD) have become invaluable tools in accelerating the materials and molecular design process.^{14–16} For gas separation and storage, atomistic grand canonical Monte Carlo (GCMC) simulations have been shown to provide relatively accurate estimates of the gas adsorption

Received: August 12, 2022

Revised: January 4, 2023

Published: January 20, 2023



properties of MOFs if the structure is known.¹⁶ The rapid increase in materials data and vast improvements in computer hardware have enabled the application of high-throughput screening and data-driven methods to further accelerate materials discovery.^{17–19} Recently, machine learning (ML) and other data-driven methods have been used to successfully develop new experimentally synthesized and validated MOFs for CO₂ capture.²⁰ To enable data-driven material discovery for any application, the first (and perhaps most crucial) step is the curation of a database. To date, tens of thousands of MOFs have been synthesized,²¹ as recorded by the Cambridge Crystallographic Data Centre (CCDC). The CCDC collates and curates the Cambridge Structural Database (CSD),²² a repository of small-molecule and metal–organic crystal structures. In 2017, the CCDC collected MOF and MOF-like structures into a separate database called the CSD MOF subset.²¹ However, the procedure used to generate this database led to the inclusion of a large fraction of MOFs which are not porous, even though permanent porosity is a key requirement for many MOF applications such as gas separation. The CSD later curated a database called the CSD MOF collection, which contains ~10,000 3D porous MOFs.²³ A smaller, yet more popular, database named the Computation-Ready Experimental (CoRE) MOF database, was created in 2014 composed of ~5000 MOFs from the CSD.²⁴ In 2016, Nazarian et al. created the “CoRE MOF 2014-DDEC” database, in which DFT-derived partial atomic charges were determined for about half of the CoRE MOF 2014 database.²⁵ The database was initially validated by using GCMC simulations of methane adsorption. An advantage this database presented over others is that it included DFT-derived charges, which enable improved accuracy of GCMC simulations when modeling adsorption of molecules where electrostatic interactions are important, such as CO₂. A subsequent study performed a DFT-level structural relaxation of the MOFs in the CoRE 2014 database to investigate the effect of solvent removal on the adsorption properties of the materials.²⁶ The CoRE 2014 database was recently expanded to ~14,000 MOFs from the CSD (CoRE 2019 database) and is currently the largest source of permanently porous experimentally characterized MOFs.²⁷ Chen and Manz screened the CoRE 2019 database to determine structures with isolated atoms, overlapping atoms, and hypo/hypervalent atoms.²⁸ Notably, the authors identified thousands of chemically invalid structures, demonstrating the requirement for a more thorough structure-checking procedure.

Databases of experimentally realized MOFs such as the CoRE and CSD databases are preferable for identifying high-performing materials since they contain synthetically feasible structures. However, the number of permanently porous MOFs suitable for gas separation is still rather limited. Thus, databases of computationally generated or hypothetical MOF (hMOF) structures have been created to explore more combinations of SBUs than that currently exist for experimentally synthesized MOFs.^{20,29–32} The approaches to generate hMOF structures can be broadly classified as either “bottom-up” or “top-down”. Bottom-up approaches involve sequentially connecting SBUs until a periodic crystal structure is obtained. Top-down approaches start with a given topology and map SBUs onto the topological net to generate a periodic crystal structure. In 2012, Wilmer et al. developed a bottom-up approach which entailed recombining 102 building blocks derived from already-synthesized MOFs.²⁹ Using predictable

assembly of building blocks into MOFs, the authors generated every possible structure within specified constraints (including geometric rules from existing MOFs), resulting in a database (“Wilmer database”) of ~138,000 hMOFs arranged in six topologies. This pioneering work resulted in the first large database of MOFs to be constructed and was initially screened for the methane uptake. Gurnani et al. recently used the Wilmer database to generate ML models to predict methane adsorption isotherms of MOFs.³³ The authors expanded the diversity of the database by substituting new metal clusters into the existing structures. Surprisingly, the model could be successfully extended to metal chemistries absent from the training set. In 2016, Boyd and Woo developed the topology-based crystal constructor (ToBasCCo) algorithm—a top-down approach which uses graph theory to generate hMOFs using only an underlying topological net, a combination of SBUs, and the bond-forming sites of the SBUs.³⁴ Importantly, this method allows for the generation of a more geometrically diverse set of structures. Using this algorithm, a MOF database (“Boyd–Woo database”) containing over 300,000 structures and over 1000 net topologies was constructed.^{20,30} Boyd et al. later screened and mined the database to identify strong, hydrophobic CO₂ binding sites which led to the experimental synthesis of water-stable MOFs for post-combustion CO₂ capture that still functioned well at high relative humidities.²⁰ Other top-down and bottom-up databases of hMOFs have also been developed, including databases where structures were created by functionalizing the pores of a parent MOF.³⁵ While hMOF databases allow for access to a wider combination of organic and inorganic SBUs, they frequently possess dramatically poorer diversity with respect to the inorganic SBUs when compared to experimental databases.^{36,37} Depending on the property of interest, using such databases could introduce undesired biases which may reduce efficiency of screening and/or transferability of ML models. Thus far, few studies have focused on correcting this lack of diversity in hMOF databases, with a notable exception being the work by Majumdar et al.,³⁶ in which new hMOFs were generated with underrepresented inorganic SBUs to expand the metal chemistry diversity of the existing hMOF space.

The computational screening of MOFs for gas separation and storage applications requires the determination of adsorption properties of the materials, which is usually performed via atomistic GCMC simulations. These simulations use a set of parameters and equations to model interatomic potentials (i.e., a “force field”). Most commonly, fixed partial atomic charges are used to model the electrostatic interactions. Although many force field parameters are generally transferable from one MOF to the next, the partial atomic charges of a specific MOF framework are not. As such, before a GCMC simulation can be performed, one must determine partial atomic charges for the MOF framework. Ideally, these charges would be derived from first-principles DFT calculations, especially when electrostatic interactions play a key role in adsorption of a particular guest molecule. The most appropriate partial atomic charges for evaluating interaction energies are so-called electrostatic potential (ESP) fitted charges where the atomic charges of the system are selected such that they optimally reproduce the ESP determined from the DFT calculation. The REPEAT method³⁸ was the first method for deriving such charges from periodic DFT calculations. However, MOFs typically have hundreds of atoms per unit cell. As such, a periodic DFT calculation to

determine the wave function and ESP can take hours on modern computing resources. Therefore, when performing a large-scale screening of a database using GCMC simulations, using DFT-derived charges can become prohibitively expensive.³⁹ This problem prompted the use of empirical charge-assignment methods, such as the charge equilibration (QEq) method⁴⁰ and the split-charge equilibration (SQE) method,⁴¹ which allow charges in MOFs to be generated within seconds. The drawback of these methods is that they must be parameterized to provide a good approximation for the ESP of the material. Several parameterizations of these methods have been developed for MOFs^{42,43} including the MEPO (MOF ESP optimized) sets^{44,45} which were fit to reproduce DFT-computed ESPs from a large training set of MOFs. The MEPO-QEq parameters have been used to generate partial atomic charges for a number high-throughput screening studies.^{46–50} While significant effort has gone into optimizing parameters for these empirical charge-assignment methods, the use of these methods on materials which deviate from structures in the original training sets will yield questionable charges.³⁹ The quantum MOF (QMOF) database⁵¹ is one notable database which contains DFT-derived partial atomic charges for ~20,000 MOFs. In a follow-up publication, Rosen et al. compare the effect of density functional on partial charges as well as other electronic properties such as band gaps.⁵² Importantly, the authors find little dependence of exchange correlation functional on partial charge assignment, concluding that the Perdew–Burke–Ernzerhof (PBE) functional is sufficient for this purpose. More recently, ML partial atomic charge models have been developed for MOFs by Raza et al.⁵³ and Kancharlapalli et al.⁵⁴ that both provide much improved results over the QEq and SQE methods. Both models were trained to reproduce DFT-derived DDEC charges on MOFs from the CoRE MOF 2014-DDEC database. In the study by Raza et al., a set of 2266 charge-labeled MOFs were partitioned into training, development, and test sets in a 70:10:20 split, while Kancharlapalli et al. used 2974 MOFs in a 20:80 split for the test and training sets, respectively. Notably, Kancharlapalli et al. removed atoms with identical chemical environments to eliminate duplicate atoms in the training and test sets. To ensure broad applicability of these ML models, it would be ideal to train ML models on large and diverse data sets.

Even with computed charge parameters available, GCMC simulations typically require millions of guest–host interaction energy calculations to determine a single point on an adsorption isotherm. As such, this type of screening falls under the category of “brute force” methods. Brute force screening of large databases of materials can range from extremely compute-intensive to intractable. Consequently, ML has been used to create models which directly relate the structure of a material to a target property (e.g., CO₂ uptake). Such models would bypass the need for a GCMC simulation altogether and allow one to compute a property for a given material at rates which are orders of magnitude faster than traditional methods. ML models can learn how to relate chemical or structural descriptors of a MOF to its performance. Descriptors should, at a minimum, uniquely characterize the material and be easier to calculate than the property itself. A descriptor should also be invariant to rotations, translations, and atom ordering. Many studies have focused on using ML to predict adsorption properties of MOFs, such as gas uptake and selectivity,^{47,55,56} adsorption isotherms,^{33,57} and Henry's constants.⁵⁸ However, one major step in these studies entails

the calculation of the descriptors, some of which are described in the [Computational Methods](#) section of this paper, and others explored in the aforementioned references as well as several research articles and reviews.^{19,59,60} An ideal database of MOFs would be “ML-ready,” with pre-computed descriptors and target data to facilitate the development of new ML models.

In this paper, we introduce a diverse database of ~280,000 experimentally characterized and hMOFs from various sources with DFT-derived partial atomic charges for GCMC screening, as well as descriptors and simulated adsorption data that can be used for developing ML models. We call this database the ab initio REPEAT charge (ARC–MOF) database, and it includes DFT-derived REPEAT charges for each MOF and descriptors that encompass three different classes, for each material in the database. These descriptor classes, which are described further in the [Computational Methods Section](#), can be generalized as geometric, atomic-property-weighted radial distribution functions (AP-RDFs),⁶¹ and revised autocorrelations (RACs)³⁷ and have been explored extensively in the literature. The RACs and geometric descriptors are then used in an in-depth diversity analysis to evaluate the chemical diversity of ARC–MOF against the current known design space of MOFs—a critical, yet commonly overlooked detail in past publications of MOF databases. Other chemical and geometric diversity analyses are presented, with an analysis on the effect of the charge-assignment method on GCMC-simulated gas uptakes in MOFs. Finally, we present a case study demonstrating the effect of the training set on ML generalizability.

■ COMPUTATIONAL METHODS

Partial Atomic Charge Calculations. *DFT-Derived Partial Atomic Charges.* Partial atomic charges were calculated from periodic DFT calculations performed on each MOF using the REPEAT method.³⁸ REPEAT fits partial atomic charges of each atom such that they collectively minimize the difference between the ESP resulting from a DFT calculation of the system and the ESP from the fitted charges. Periodic DFT calculations were performed on all MOFs using VASP version 5.4.4^{62,63} with the PBE functional⁶⁴ and the projector augmented-wave method.⁶⁵ All DFT calculations performed in this work consumed a total of approximately 2200 CPU years on recent generation Intel processors. A $3 \times 3 \times 3$ Monkhorst–Pack sampling of k-points in the Brillouin zone was used for all MOFs whose smallest cell vector length is less than 14 Å. For MOFs whose cell vectors were all greater than 14 Å, only the Γ -point was sampled. A random sampling of 3000 MOFs whose minimum cell-vector length was between 13.5 and 14.0 Å showed that the REPEAT charges calculated with a Γ -point sampling had a mean absolute deviation (MAD) from those determined with a $3 \times 3 \times 3$ sampling of k-points of only 0.0024 e. The same comparison on a random sampling of 3000 MOFs whose minimum cell-vector length was less than 7 Å was found to have a MAD of 0.0156 e, which is an order of magnitude higher. Since insufficient Brillouin zone sampling is more acute for smaller cells, this shows that our choice of sampling only the Γ -point for MOFs with cell vectors larger than 14 Å is justified. The VASP calculations used a plane wave cutoff of 300 eV and a lower than default FFT grid density (VASP keyword “PREC = low”) to reduce computing time. To justify this choice of lower precision settings for the DFT calculations, we have calculated the REPEAT charges for a set of MOFs using a 400 eV plane wave cutoff, a “PREC = normal” setting, and a $6 \times 6 \times 6$ Monkhorst–Pack sampling of k-points. The REPEAT charges derived from the higher precision calculations were then compared to the charges calculated using the low precision settings (PREC = low and a $3 \times 3 \times 3$ sampling of k-points for MOFs possessing a cell vector less than 14 Å and Γ -point sampling otherwise). For a diverse set of 111 MOFs taken from various

databases, with 16 different metals (Cu, Ni, Fe, Mo, Cd, Zn, V, Co, Mg, Mn, In, B, Zr, Ag, Al, and La) and at least 12 different topologies, the difference in REPEAT charges using the two precision settings was found to have a MAD of only 0.0032 e over 10,572 atomic centers. Furthermore, the average maximum absolute deviation over all MOFs was only found to be 0.015 e. For the REPEAT calculations, no restraints were used, but the default van der Waals radii were scaled by a factor of 0.9.

Only MOFs that were expected to have neutral frameworks were selected from the various sources for ARC–MOF. This was carried out by predicting the oxidation state of metals in each MOF assuming a neutral framework. MOFs with metals containing an invalid oxidation state in the neutral framework were removed. Additionally, most MOFs were assumed to be closed-shell and calculated using a restricted-DFT formalism. Any MOFs containing the elements Cr, Mn, Fe, Co, Ni, Gd, Dy, Ho, and Er that commonly have unpaired electrons were calculated using an unrestricted-DFT formalism.

Empirical Partial Atomic Charges. To determine the effect of charge-assignment method on adsorption properties of MOFs, the commonly used empirical MEPO–QE⁴⁴ method was compared to the charges obtained by REPEAT.

Structure Validation. To ensure that ARC–MOF contains only chemically reasonable structures, the oxidation state of each metal atom in each structure was determined. The code used to do this is available in the [Supporting Information](#). Structures containing metals with impossible, unknown, and/or non-integer oxidation states were excluded from this study. Structures which possessed atom-pair distances of less than 70% of the sum of their covalent bond radii were discarded to avoid including structures which possess overlapping atoms. Lattice parameters were also checked to avoid structures with unrealistically small cell vectors. The smallest cell vector in the CSD MOF collection, which only contains experimentally synthesized MOFs, was found to be 3.22 Å.⁶⁶ Therefore, only MOFs with unit cell vectors greater than this length were included in ARC–MOF. Finally, structures with hypercoordinated main-group elements were discarded. For this check, structure graphs were generated (as described in the [Chemical Substructure Identification](#) methodology) for each MOF with the metal atoms removed since metals frequently involve hypercoordinated oxygen and hydrogen. Then, MOFs containing carbon atoms with more than four bonds, oxygen atoms with more than two bonds, and halogen/hydrogen atoms with more than one bond were discarded. For experimental MOFs containing counterions, perchlorate and hydronium ions were excluded from this check.

Chemical Substructure Identification. Trends in the incidence of various chemical substructures (i.e., chemical moieties and organic functional groups) in ARC–MOF were determined using bond connectivity data in the crystallographic information files (CIFs) of the structures. For structures which had CIF files missing bond connectivity tables, the requisite bond connectivity tables were generated using the nearest-neighbor algorithm proposed by Isayev et al.,⁶⁷ as implemented in the Python cheminformatics library pymatgen (version 2022.0.5).⁶⁸ In this nearest-neighbor algorithm, adjacent atoms are considered bonded if they shared a Voronoi facet, and the distance between them was less than their summed Cordero covalent radii⁶⁹ plus an additional bond tolerance value. A bond tolerance value of 0.25 Å was selected to represent only strong interatomic interactions⁶⁹ and to limit hypercoordination at carbon atom and metal atom sites. The substructures were extracted by identifying bonded atom pairs within its connectivity table and then evaluating each atom's first coordination sphere. Only substructures not bound to metal atoms were considered in this analysis. These fragments were compared against a representative library of chemical substructures to identify which ones were present within each ARC–MOF structure.

Topology Identification. The Julia package CrystalNets⁷⁰ was used to determine the topology of the MOFs in ARC–MOF, using the bonding tables obtained from pymatgen (as described in the [Chemical Substructure Identification](#) methodology). The auto clustering option with the MOF structure type was used. The “detect_organiccycles” option was set to false to improve the identification accuracy. Topology information is available a priori

from the filenames of 207,453 MOFs in ARC–MOF, but the remaining MOFs have no indication of topology. Therefore, the CrystalNets result was used for the remaining 72,257 MOFs. The CrystalNets result matched the filename of the MOFs 93% of the time, suggesting a reasonable prediction accuracy. CrystalNets failed to identify a topology for 15,349 MOFs, so topological data are only available for 264,262 MOFs in ARC–MOF (95%). The predicted topologies of the MOFs are available in the [Supporting Information](#).

Descriptor Calculations. Geometric Descriptors. Geometric descriptors of all materials were calculated with Zeo++⁷¹ version 0.3.0 with the high accuracy flag and a probe radius of 1.86 Å. The accessible surface area, accessible volume, and probe-occupiable volume were calculated with 2000, 50000, and 2000 MC steps, respectively. A list of the 20 geometric properties evaluated for each MOF is given in the [Supporting Information](#).

Revised Autocorrelation Function Descriptors. RACs are provided as descriptors for ML studies and are additionally used in this work to evaluate the diversity of our database in comparison to other MOF databases. The RACs were computed according to the methodology described by Moosavi et al.³⁷ RACs are products or differences of properties computed on a non-weighted chemical graph. An example of a difference-based RAC is

$$p_{d, \text{scope}}^{\text{diff}} = \sum_i^{\text{start}} \sum_j^{\text{scope}} (P_i - P_j) \delta(d_{ij}, d)$$

where d is the “depth” of the RAC, equal to the bond-wise path distance from the starting atom, and d_{ij} is the bond-wise path distance between atoms i and j . δ is the Kronecker delta, equal to unity when d_{ij} is equal to d , and equal to zero otherwise. P_i and P_j represent atomic properties of atom i and j , respectively. The start and scope lists for each type of RAC and the RACs included in this work are given in the [Supporting Information](#). The properties considered for the RACs in this work are electronegativity, nuclear charge, atom identity, connectivity, and covalent radii. For the linker connecting atom and functional group RACs, an additional property (polarizability) is included. A single descriptor value is generated for each property and depth, and these values are stacked to form a single feature vector to describe a particular type of chemistry of a MOF. To compute the RACs, the MolSimplify code (version 1.5.0) written by Kulik and coworkers was used.⁷² The start and scope atom lists used in this work are given in [Table S2](#).

Atomic-Property-Weighted Radial Distribution Function Descriptors. An in-depth definition of AP-RDF descriptors⁶¹ and their variants is given elsewhere.⁴⁷ Briefly, the AP-RDF descriptor gives an atomic property weighted probability of finding atom pairs separated by a given distance inside a unit cell of a MOF. In this work, we define the AP-RDF descriptor using the following equation

$$\text{RDF}^P(R) = f \sum_{i,j}^{\text{atom pairs}} P_i P_j e^{-B(r_{ij}-R)^2}$$

where r_{ij} is the minimum image distance between atoms i and j , P_i and P_j are atomic properties of atom i and j , respectively, B is a Gaussian smoothing parameter, and f is a normalization factor. In previous work,⁶¹ the optimal value of B was found to be 10. Each AP-RDF was normalized by the number of framework atoms in the unit cell (N_{atoms}) and scaled by a factor of 0.001 (i.e., $f = 0.001/N_{\text{atoms}}$). The AP-RDF descriptors were calculated using a distance range of 2.0–30.0 Å in 113 steps that linearly increase from 0.004425 to 0.5 Å, for the following atomic properties: electronegativity, atomic hardness, van der Waals volume, dipole polarizability, atomic mass, and none (i.e., P_i and $P_j = 1$), yielding a total of 678 AP-RDF descriptor values for each MOF. The electronegativity parameter was used, as described by Cottrell and Sutton, computed as the square root of the quotient of the effective nuclear charge and covalent radius.⁷³ The hardness was computed as the difference in ionization energy and electron affinity divided by two. Notably, electron affinity is not available for all elements of the periodic table, and consequently, the AP-RDF with the hardness property is not available for all MOFs in ARC–MOF.

Table 1. Labeling Used to Identify Sources of Structures to Construct the ARC–MOF Database, Compositions, the Number of MOFs Remaining after the Structure Checking Procedure, and the Number of MOFs Present from Each Database in ARC–MOF^a

label	references	database composition				total no. of MOFs	no. of MOFs after structure check	no. of MOFs in ARC–MOF (%)
		no. organic SBUs	no. inorganic SBUs	no. functional groups	no. topologies			
DB0	Boyd et al. ^{20,34}	175	23	50	1166	358,398	263,218	203,025 (72.2)
DB1	Lan et al. ³²	32	17	9	18	303,992	181,885	23,267 (8.3)
DB2	Colón et al. ³¹	60	14	0	41	13,514	3920	199 (0.1)
DB3	Anderson et al. ⁹²	12	4	8	15	426	358	123 (0.0)
DB4	Gómez-Gauldrón et al. ⁹³	48	2		4	204	48	25 (0.0)
DB5	Chung et al. ⁹⁵	82	5	12	6	51,163	27,022	22,366 (8.0)
DB6	Li et al. ⁹⁶	41	1	3	1	11,555	10,944	9092 (3.2)
DB7	Majumdar et al. ³⁶	95	14		52	23,891	12,316	6955 (2.5)
DB8	Anderson and Gómez-Gauldrón ¹⁰⁰	7	1		19	126	122	8 (0.0)
DB10	Anderson et al. ⁹⁷	5	4		17	105	78	22 (0.0)
DB12	CoRE 2019 ²⁷		15,613	8379	7145 (2.5)			
DB13	Bao et al. ⁹⁸	100	5		9	8629	6180	5165 (1.8)
DB14	CSD MOF ²³		10,636	4070	72 (0.0)			
DB15	this work	195	19		37	7708	2841	2146 (0.8)
total						806,520	521,381	279,610

^aThe percentages in parentheses represent the percentage contribution of each database to ARC–MOF. Information on database composition is given only for hMOF databases.

Diversity Analysis. The analysis presented herein was accomplished by expanding upon the methodology proposed by Moosavi et al.³⁷ The chemistry-specific RAC descriptors for (a) metal-center chemistry; (b) ligand chemistry; and (c) functional group chemistry, as well as the six geometric descriptors, as shown in Table 2, were employed in this analysis. The atomic properties used to compute the RACs in this work are provided in the Supporting Information. RACs are graph-based descriptors which describe the chemistry of the various “building blocks” of MOFs. In previous work, a depth of two was used to describe the metal chemistry.³⁷ In our opinion, this is too local to the metal as there can be some chemically distinct local metal environments if one considers a depth greater than two. An example of this is a metal bound to a substituted versus unsubstituted pyridine ligand. For these reasons, a depth of three was chosen for all RAC descriptors used in this analysis. The uniform manifold approximation and projection (UMAP) technique,⁷⁴ as implemented in the RAPIDS cuML Python library,⁷⁵ was used to reduce the high-dimensional data for each type of descriptors to only two dimensions for visualization purposes (e.g., the 20-dimensional metal-center RAC descriptors were reduced to two dimensions using UMAP). Each set of descriptors was stacked into a feature vector as the input for the dimensionality reduction without any scaling. Hyperparameters used for the generation of these plots are given in the Supporting Information. In this work, we define the “entire” design space as the collection of all MOFs taken from the sources listed in Table 1 that passed our structure checking protocols (~521 K MOFs). ARC–MOF is a subset of this design space. Structures for which an appropriate periodic graph could not be constructed by MolSimplify were not included in the analysis since RACs could not be computed for these structures (~9% of the overall space—49,402 MOFs). Thus, 480 K MOFs were used in this analysis to represent the “entire” design space. Due to practical limitations of using UMAP on large data sets, a random set of 50 K MOFs was selected to represent the entire set of 480 K MOFs (~10% of the entire data set) for the UMAP plots. While these plots allow for qualitative analysis of the data, quantifiable metrics are desirable. For this purpose, three metrics used by Moosavi et al.³⁷ were employed in this analysis. The first metric, disparity (*D*), quantifies the spread of the data compared to another data set. Thus, this metric uses the UMAP plots to obtain a ratio between the 2D area occupied by the colored points (ARC–MOF) and the 2D area occupied by all the points (entire design space). This metric is equal

to unity for a subset which spans the exact same space as the overall set. The other two metrics, variety (*V*) and balance (*B*), depend on the determination of clusters. HDBSCAN,⁷⁶ as implemented in the hdbscan Python library,⁷⁷ was used to cluster the MOFs based on RAC descriptors (see the Supporting Information for the clusters of MOFs). Consequently, these cluster-based metrics are independent of the UMAP plots (unlike the disparity). A previous comparison of dimensionality reduction techniques for the calculation of disparity suggests that graph-based dimensionality reduction techniques (such as UMAP) should yield similar results compared to matrix-factorization methods like PCA, which is supported by some initial tests in this paper.³⁷ While t-distributed stochastic neighbor embedding (t-SNE) was used in previous studies, UMAP has been used in this paper since it has been demonstrated to preserve more global structure than t-SNE while still preserving the local structure for some data sets.⁷⁴ Therefore, UMAP should exhibit higher fidelity between the high-dimensional and dimension-reduced spaces. Variety is a measure of how many clusters are occupied by the subset versus the number of total clusters and is equal to unity for a subset which has at least one structure in each cluster. Finally, the balance is computed using Pielou’s evenness, a well-behaved transformation of Shannon entropy, which is equal to unity for a completely balanced data set. Therefore, this is the only absolute metric (i.e., does not involve a comparison of the subset to the total set). Detailed information on the calculation of these metrics, as well as the code to compute these metrics, can be found in the Supporting Information. Farthest point sampling⁷⁸ (also known as max/min sampling) was used to select a more balanced set of structures for each unbalanced type of chemistry in ARC–MOF (i.e., functional group chemistry and metal chemistry). Balanced subsets of ARC–MOF are provided in the Supporting Information, as well as code to perform the farthest point sampling.

Grand Canonical Monte Carlo Simulations. GCMC simulations were performed using an in-house code based on the DL_POLY MD package to calculate equilibrium gas adsorption properties. Guest-framework interactions were modeled using Lennard-Jones (LJ) potentials to account for steric and dispersion interactions, and the fixed partial atomic charge model was used to model non-bonded electrostatic interactions. The LJ parameters for the frameworks were assigned from the UFF force field. The REPEAT and QEq methods were used to assign partial charges to the

framework atoms to allow for comparison between the charge-assignment methods. Lorenz-Berthelot mixing rules were used for the determination of LJ parameters between atoms of different types. The intermolecular potential parameters for CO₂ were taken from a study by Garcia-Sánchez et al.⁷⁹ The parameters for H₂S were taken from Kamath et al.⁸⁰ The parameters for N₂ were developed in-house to reproduce experimental N₂ uptake isotherms in MOFs.⁸¹ The parameters for CH₄ were taken from a study by Stubbs et al.⁸² Finally, the parameters for H₂ were taken from a study by Belof et al.⁸³ For the GCMC simulations used in comparison of charge-assignment methods, 200,000 MC steps were performed, split evenly between equilibration and production. 10,000 MOFs were selected to perform the charge-assignment comparison. Additionally, GCMC screening of ARC–MOF was performed to obtain the gas uptake relevant to five different gas separations. For these simulations, 10,000 MC cycles were performed, split evenly between equilibration and production, and the REPEAT charges reported in this work were used for the partial charges of the framework atoms. A single MC cycle is defined in this work as *X* steps, where *X* is the number of guests in the simulation cell at the beginning of the cycle. The gas uptake heat of adsorption data, as well as working capacity and selectivity for these gas separation conditions are given in the [Supporting Information](#) and intended for use as target properties for future ML studies. For the uptakes, error estimates are also provided as the standard deviation over five windowed averages of the properties over the production phase of the simulation. The following processes were used to determine the relevant gas separation conditions given in the [Supporting Information](#): (a) methane purification, (b) post-combustion VSA, (c) pre-combustion PSA, (d) methane storage PSA, and (e) landfill gas VPSA.

Gradient Boosted Decision Tree Regression Models.

Gradient boosted decision tree (GBDT) regressors were used to compare the impact of training set on the ML performance. The MC model is defined as the model trained using the MOFs in the diverse-metal subset which are present in ARC–MOF (~12,000 MOFs, ~34,000 metal charges). The ARC model is defined as the model trained using a 5% random selection of MOFs from ARC–MOF (~14,000 MOFs, ~34,000 metal charges). For each training set, a random split of 95:5 train/development was used to preserve as much training data as possible. CatBoost⁸⁴ was used to train the GBDTs (depth of 10 and mean absolute error (MAE) as the loss function) with the plain boosting type since this improved the prediction performance on the development set. Smooth overlap of atomic position (SOAP) descriptors⁸⁵ were calculated, as implemented in version 0.3 of the DScribe Python library⁸⁶ to compute the input features corresponding to each metal atom. For the SOAP vectors, a cutoff radius of 6.0 Å, $n_{\text{max}} = 4$, and $l_{\text{max}} = 3$ were used. Notably, since some cells contain multiple crystallographically identical metal sites, only metal sites with distinct SOAP vectors were added to the training set to prevent the inclusion of redundant data. The similarity of SOAP vectors was computed using the SOAP similarity kernel derived by De et al.⁸⁷ with a distance cutoff of 0.01. Since the size of the SOAP vectors depends on the number of unique elements in each crystal structure, and since they are high-dimensional (dimension of up to ~3000), post-processing of these vectors was necessary to generate suitable inputs for the ML models. Thus, the SOAP vectors were first zero-padded to produce vectors of equal length.⁸⁶ Then, a scaler model and PCA model were fit to the zero-padded SOAP vector of each metal atom in ARC–MOF. Each SOAP vector was transformed using this scaler and PCA model. The first 300 principal components were extracted from each vector to produce a single 300-dimensional vector per MOF as the input for the GBDT models. The scaler used was StandardScaler from the Scikit-Learn Python library,⁸⁸ and the PCA model was fit using the RAPIDS CuML Python library.⁸⁹ To evaluate the performance of these ML models, a test set of MOFs from the diverse-metal set which are not in ARC–MOF (~7000 MOFs) was used. REPEAT charges were computed for this set of MOFs following the methodology outlined in this paper. This test set is provided in the [Supporting Information](#) as CIF files with partial atomic charges.

RESULTS AND DISCUSSION

ARC–MOF Composition. ARC–MOF contains both experimentally characterized and hMOF structures taken from several different sources. Notably, the structures in the present study have not been modified if taken from third party databases. For example, DFT level optimizations were not performed on the structures. In the case of DB15, which are new structures created for this work, the structures were optimized with the UFF force field. For simplicity, we will refer to each source as a database and have assigned each a unique label—DB*x* where “*x*” is an identifier number starting from zero. [Table 1](#) summarizes the composition of each database used to curate ARC–MOF, including the number of organic SBUs, inorganic SBUs, functional groups, and topologies used to construct the MOFs in each database, if available. The total number of MOFs from each database is given, with the number of MOFs remaining after structure checking, and the number of MOFs from each database present in ARC–MOF. CIF files of all structures in ARC–MOF are available (see [Supporting Information](#)), with the filename comprising the “DB*x*” prefix from [Table 1](#) followed by the name of the structure given in the original source. All of the structures considered in this work which passed our structure checking are also included in the [Supporting Information](#). DB0 represents the hypothetical database previously developed by our group, referred to in the literature as the Boyd–Woo database.²⁰ This is a database constructed from both the top-down and bottom-up MOF construction methodologies. However, most structures in the database were made using the top-down ToBasCCo code developed by Boyd and Woo.³⁴ All structures in the Boyd–Woo database were optimized with the UFF force field⁹⁰ where the geometry of the metal and atoms directly bonded to the metal were frozen. DB1 is an hMOF database constructed by Lan et al.³² using their top-down crystal construction algorithm⁹¹ to generate hypothetical ionic liquid/MOF composites. DB2 was created by Colón et al.³¹ using their top-down MOF builder code topologically based on the crystal constructor (ToBaCCo)³¹ and optimized the structures using molecular mechanics. DB3 contains hMOFs generated using ToBaCCo by Anderson et al.,⁹² optimized using UFF. The organic SBUs used in DB3 consist only of functionalized benzene dicarboxylate ligands, with the following functional groups: amino, bromo, fluoro, hydroxyl, nitro, methyl, thiol, and trifluoromethyl groups. DB4, reported by Gómez-Gualdrón et al.,⁹³ consists of hMOFs featuring Zr₆O₄(OH)₄(CO₂)₁₂ and Zr₆O₄(OH)₄(CO₂)₈(OH)₄ SBUs. 12 ditopic organic SBUs were combined with the inorganic SBUs to create MOFs with the *fcu* topology, and 36 tetratopic organic SBUs were combined with the inorganic SBUs to create MOFs with *ftw*, *csq*, and *scu* topologies. The MOFs were built and optimized in Materials Studio⁹⁴ using the Crystal Builder and Forcite modules, respectively. DB5 is a subset of the Wilmer database²⁹ containing only MOFs with a unique combination of interpenetration capacity, actual interpenetration level, inorganic node, primary linker, secondary linker, and functional groups, as curated by Chung et al.⁹⁵ DB6 contains hMOFs based on the Cu-paddlewheel SBU possessing *pcu* topology.⁹⁶ An additional 560 unfunctionalized parent versions of the MOFs were also generated. DB7 is an hMOF database generated by Majumdar et al., with the focus of using inorganic SBUs not commonly observed in hMOF databases.³⁶ DB8 is a set of hMOFs reported by Anderson and Gómez-Gualdrón, in

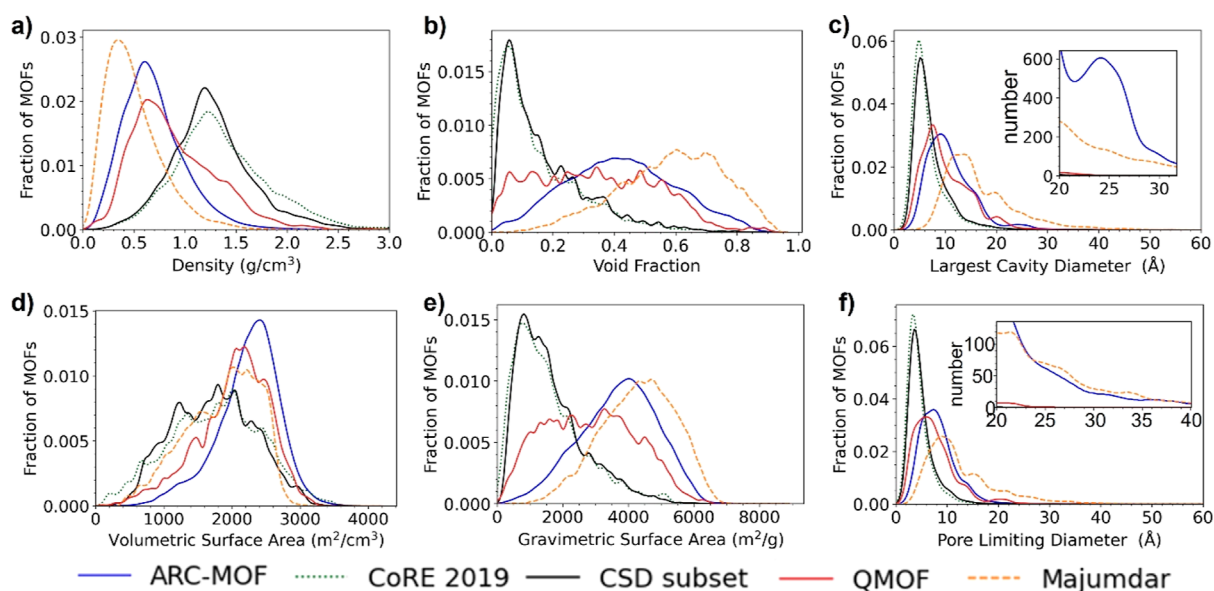


Figure 1. Distribution of common geometric parameters: (a) density; (b) void fraction; (c) largest cavity diameter; (d) volumetric surface area; (e) gravimetric surface area; and (f) pore limiting diameter within the ARC–MOF database (solid blue), the full CoRE 2019 database²⁷ (dotted green), the CSD MOF database²³ (solid black), the QMOF database⁵¹ (solid red), and the Majumdar database³⁶ (dashed orange). The insets in (c,f) show the absolute number of MOFs. The fraction of structures with zero-surface areas, void fractions, or diameters is not plotted, although they are included in the calculation of the distributions.

Table 2. Maxima and Minima of Selected Geometric Properties for ARC–MOF and Other MOF Databases

database		density (g/cm ³)	volumetric surface area (m ² /cm ³)	gravimetric surface area (m ² /g)	volume fraction	largest cavity diameter (Å)	pore limiting diameter (Å)
ARC–MOF	max	6.20	3475	10,219	0.95	83.1	81.1
	min	0.02	0	0	0	1.6	0.1
CoRE	max	4.16	3150	8309	0.89	71.6	71.5
	min	0.06	0	0	0	2.7	1.1
CSD	max	4.06	3153	6621	0.80	71.6	71.5
	min	0.13	0	0	0	2.7	0.5
QMOF	max	2.88	2878	7438	0.88	33.7	30.1
	min	0.08	0	0	0	1.9	0.9
Majumdar	max	2.09	2724	8013	0.92	56.6	54.2
	min	0.05	0	0	0	4.1	3.0

which the Cu-paddlewheel SBU was combined with tetratopic organic SBUs using a modified version of ToBaCCo. The resulting structures were optimized in LAMMPS using the UFF4MOF force field. DB10 contains hMOFs created using ToBaCCo and optimized using UFF.⁹⁷ DB12 is the CoRE 2019 database.²⁷ Bao et al. created DB13, which contains MOFs comprising organic SBUs evolved using a genetic algorithm to maximize deliverable methane capacity from an initial population of 57,815 commercially available molecules.⁹⁸ DB14 is the CSD–MOF collection, which contains structures from the non-disordered subset of the CSD.²³ Only structures with the H-atoms added were sampled from DB14, and structures with the same CSD code that were included from the CoRE database were not added to ARC–MOF to avoid duplicate structures. DB15 refers to new structures created for this work using a top-down approach, as implemented in Pormake.⁹⁹ To sample MOFs from each database, the structures were sorted based on the number of atoms in the unit cell. DFT calculations were first calculated on the smallest members of each database moving to larger structures incrementally until the DFT calculations were no longer feasible with the computing resources available. For

DB1, which contains over 300 K MOFs, the above procedure was followed on a random subset of ~75,000 structures. Structures where the wave function would not properly converge were discarded. Only MOFs which had a valid metal oxidation state in a neutral framework were selected, to ensure that all MOFs in ARC–MOF can be treated as neutral.

CHEMICAL AND GEOMETRIC DIVERSITY OF ARC–MOF

We now examine both the chemical and geometric diversity of ARC–MOF, with a comparison to other databases of MOFs. Figure 1 shows the distribution of common geometric descriptors within ARC–MOF in comparison to the CoRE 2019,²⁷ CSD–MOF,²³ QMOF,⁵¹ and Majumdar³⁶ databases. The distributions plotted are for the entire databases. Since it is difficult to determine the maxima and minima of the geometric parameters from the plots shown in Figure 1, these are given in Table 2 for each database.

Examination of the plots in Figure 1 reveals that the databases containing hMOFs (QMOF, Majumdar, and ARC–MOF) have distributions of geometric parameters that are mutually similar compared to purely experimental MOF

databases (CoRE and CSD–MOF). The hypothetical databases have distributions that are more skewed toward structures with larger pore diameters, surface areas, and void fractions, indicating that they contain materials that are more porous than the two purely experimental databases. This is consistent with the fact that the experimental MOF databases are also higher in density, on average. It would be prudent to specifically compare the ARC–MOF and QMOF databases since DFT calculations have been performed on all materials in each. In general, the two databases have similar distributions for the six geometric parameters, as shown in Figure 1. Perhaps the most apparent difference is that ARC–MOF has a more normal distribution of void fractions and gravimetric surface areas, as compared to QMOF which has more uniform distributions for these parameters in the low value regimes. Close inspection of the distribution of largest cavity diameter and pore limiting diameter might suggest that the QMOF database has more structures with larger pores. However, it is important to realize that ARC–MOF is a much larger database than QMOF. The inset plots in Figure 1c,f show the absolute numbers of MOFs for those parameters, rather than the fraction (Figure S1 provides plots of the absolute values for all six geometric parameters). This reveals that ARC–MOF has a greater number of MOFs over the ranges of large diameters than QMOF. Additionally, Table 2 shows that ARC–MOF has maximum diameters that are more than twice that of QMOF.

A second analysis that was performed is related to the chemical substructures present in ARC–MOF, as summarized in Figure 2. The most common moieties were found to be

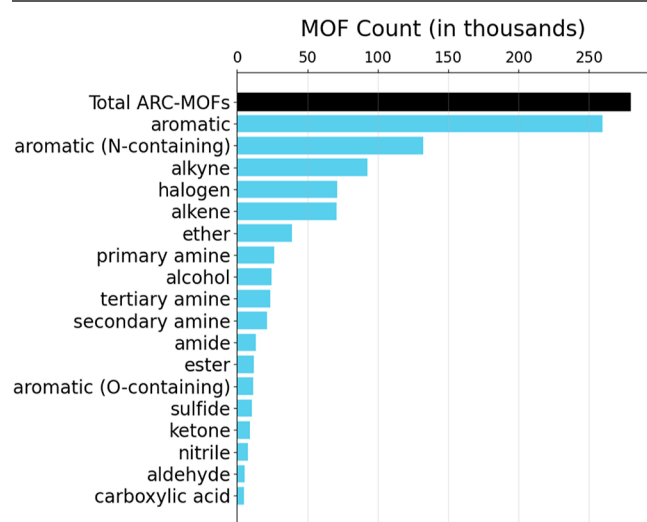


Figure 2. Comparison between the incidence of chemical substructures in ARC–MOF. Only substructures not bound to metal atoms were considered in this analysis.

aromatics (93.0% of all MOFs with 47.4% containing an *N*-heterocyclic aromatic), alkynes (33.1%), alkenes (25.3%), halogens (25.5%), and amines (24.2%). Other significant functional groups—including acids, alcohols, amides, sulfides, and so forth—were also present in the database structures in relatively smaller quantities. The results of our database functional group analysis appear consistent with the moieties most frequently observed in organic linker SBUs, employed in MOF syntheses and those included in hypothetical SBU libraries.

Topological data (i.e., information about the node-linker connectivity and symmetry) is available in the filenames of over 210,000 structures in the ARC–MOF database. For the MOFs for which the topology is unknown, we have used the Julia package CrystalNets to determine them where possible (~95% of ARC–MOF). Of these MOFs, the frequency of the 20 most common net topologies in ARC–MOF are summarized in Table 3, with the raw data given in the

Table 3. Top 20 Most Common Net Topologies in ARC–MOF, the Count, and Percentage of MOFs in ARC–MOF with a Given Topology, Respectively

topology	count	%
<i>pcu</i>	100,482	38.0
<i>fsc</i>	45,494	17.2
<i>nbo</i>	43,087	16.3
<i>pts</i>	37,193	14.1
<i>sra</i>	10,629	4.0
<i>rna</i>	4206	1.6
<i>dia</i>	3859	1.5
<i>tfzd</i>	2980	1.1
<i>cds</i>	1615	0.6
<i>lvt</i>	1330	0.5
<i>sqc</i>	873	0.3
<i>tbo</i>	813	0.3
<i>bcu</i>	807	0.3
<i>sta</i>	786	0.3
<i>nou</i>	753	0.3
<i>fof</i>	691	0.3
<i>hxg</i>	510	0.2
<i>pto</i>	362	0.1
<i>fcu</i>	319	0.1
<i>tfz</i>	272	0.1
other	6923	2.6

Supporting Information. The majority of MOFs in ARC–MOF are identified as possessing *pcu* (38.0%), *fsc* (17.2%), *nbo* (16.3%), or *pts* (14.1%) net topologies. However, the database consists of 494 unique net topologies, with 46 topologies being represented in a minimum of 50 structures.

Biases are frequently introduced in the curation of any MOF database which can affect the transferability of ML models and efficiency of large-scale screening. For these reasons, the importance of the chemical diversity of MOF data set in any high-throughput workflow cannot be overstated. For example, Moosavi et al. previously demonstrated that there is a substantially lower diversity of the metal chemistry of hMOFs compared to the design space covered by experimental databases.³⁷ Therefore, as a final step in investigating the diversity of the structures in ARC–MOF, an in-depth analysis using RAC descriptors and geometric descriptors was performed, as described in the Computational Methods Section. In this work, we define the “entire” design space as the collection of all MOFs taken from the sources listed in Table 1 that passed our structure checking protocols (~521 K MOFs), as outlined in the Computational Methods Section. ARC–MOF is a subset of this design space. The chemistry-specific RAC descriptors for (a) metal-center chemistry, (b) ligand chemistry, and (c) functional group chemistry, as well as the six geometric descriptors shown in Table 2 were used in this analysis. Structures for which a periodic graph could not be constructed were not included in the analysis since RACs

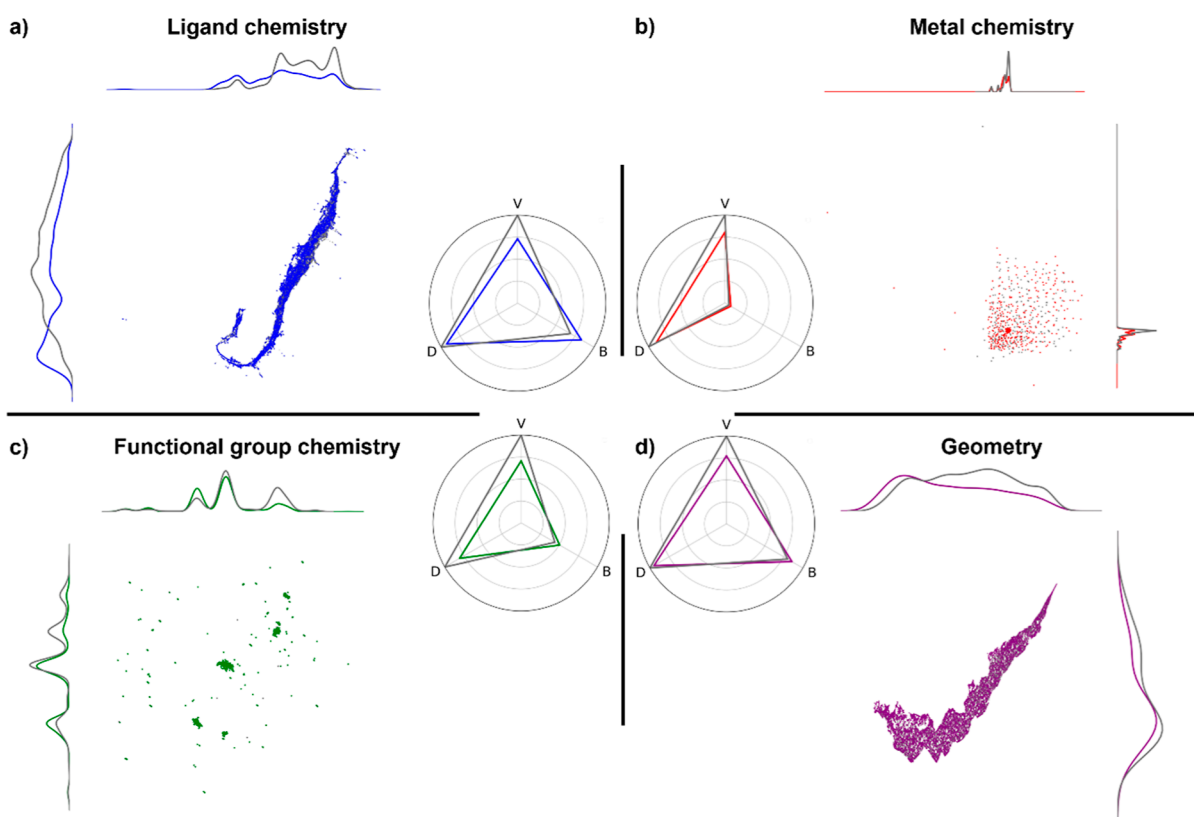


Figure 3. Two-dimensional UMAP projection of descriptors of 50,000 random MOFs and corresponding radar plots showing diversity metrics for the (a) ligand chemistry, (b) metal chemistry, (c) functional group chemistry, and (d) geometry. RAC descriptors were used for the chemical diversity (a–c) and geometric descriptors were used for the geometry (d). Structures present in ARC–MOF (~280 K MOFs) are represented by colored points, which is overlaid on the entire design space (~480 K MOFs), represented by gray points. The diversity metrics shown on the radar plots are variety (*V*), disparity (*D*), and balance (*B*), where gray again represents the entire set of MOFs and color represents the ARC–MOF subset. Only functionalized MOFs were considered for (c), and only MOFs with non-zero accessible surface area are shown in (d).

could not be computed for these structures (~9% of the overall space—49,402 MOFs). Thus, 480 K MOFs were used in the following analysis. While we regard this as an inherent limitation of using RACs for this analysis, these are presently the best descriptors available in the literature for this purpose.

Ideally, one would aim to construct a scatterplot of these descriptors and visualize a “map” of the chemistry of the design space, as described by the RACs, and thereby analyze the chemical diversity. However, since the RACs are a high-dimensional representation of the design space, one must reduce this space to a two- or three-dimensional space to allow for visualization. For this task, UMAP was used to display the high-dimensional data in a two-dimensional space for each type of descriptor to generate so-called UMAP plots (Figure 3). Due to the practical limitations of using UMAP on large data sets, a random set of 50 K MOFs was selected to represent the entire set of 480 K MOFs (~10% of the entire data set). The UMAP plots show how well the subset (ARC–MOF) covers the entire known design space, and how uniformly this space is mapped (i.e., whether clusters exist in the data). On the two axes, kernel density estimate plots of the UMAP data for each dimension show the density of the points in the space, giving a qualitative measure of the balance of the dimension-reduced data. The radar plots in Figure 3 show the value of each diversity metric (disparity, variety, and balance), which are described in the Computational Methods Section. As shown in Figure 3, colored points on the UMAP plots and

colored lines on the radar plots represent ARC–MOF, while gray points and lines represent the entire design space.

Figure 3 suggests that ARC–MOF is sufficiently diverse with respect to the presently known design space. This is perhaps an unsurprising conclusion given ARC–MOF is a near random sample of 54% of this space and as such should have approximately equal diversity compared to the overall space. Clearly, the absence of substantial gray regions in Figure 3 suggests that ARC–MOF covers the same design space as the overall set of MOFs (this is reflected in the disparity metric). The variety and balance are also approximately equal between the two sets. However, while the balance of the geometry and ligand chemistry of ARC–MOF are almost equal to unity (suggesting a nearly uniform distribution), this is not the case for the metal chemistry and functional group chemistry. Since the balance and variety calculations are independent of the UMAP plots in Figure 3, clustering was carried out on the entire data set, rather than the random 50 K MOF sample, to achieve a more accurate determination of balance of ARC–MOF. The results of this analysis are summarized in Table 4. Table 4 shows a very low balance of 0.01 for the metal-center RACs in both ARC–MOF and the entire design space, in agreement with the results of Figure 3b (Table 4). The imbalance of metal chemistry in ARC–MOF (and the overall design space) is expected as most MOF structures in the design space are hypothetical (i.e., computationally generated). In hypothetical database generation, there is typically a bias toward small number (usually in the tens or even less) of

Table 4. Balance and Variety of the Design Space (~480 K MOFs) and ARC-MOF (~280 K MOFs)

type of MOF diversity		balance	variety
ligand chemistry	all	0.55	1.00
	ARC-MOF	0.78	0.74
metal chemistry	all	0.01	1.00
	ARC-MOF	0.01	0.76
functional group chemistry	all	0.20	1.00
	ARC-MOF	0.24	0.76
Geometry	all	0.66	1.00
	ARC-MOF	0.83	0.74

inorganic SBUs that are used to produce the structures, but hundreds of inorganic SBUs have been experimentally incorporated into MOFs and are found in ARC-MOF. Although the balance of the functional group chemistry within ARC-MOF (and the entire design space) is significantly higher than that of the metal chemistry, it is still not particularly high with a value of only 0.24 in ARC-MOF and 0.20 in the entire design space (Table 4).

Two common ways to address data imbalance are to either add duplicate/artificial data to the underrepresented clusters (upsampling) or remove data from the overrepresented clusters (downsampling). In this work, farthest point sampling⁷⁸ (also known as max/min sampling) was used to select a more balanced set of structures for each unbalanced

type of chemistry in ARC-MOF (i.e., functional group chemistry and metal chemistry). Notably, the RACs used for the farthest point sampling depended on the type of chemistry (e.g., only the metal-center RACs were used for sampling the MOFs with most diverse-metal chemistry). Farthest point sampling is a greedy algorithm which iteratively samples points from a set, such that the sampled points optimally cover the data set (i.e., they are “spread out”). Using this technique, in combination with a clustering technique (for unclassified data), one can determine a subset which has maximum size and balance for a particular type of chemistry. The diversity metrics and UMAP plots for these balanced subsets are shown in Figure 4. For the ligand chemistry and geometry, a farthest-point-sampled set of 100,000 MOFs was used to demonstrate that the diversity was unaffected since these types of RACs are already well-balanced over the entire set. It was determined from this analysis that a subset of 150,000 MOFs was reasonable to obtain a set of MOFs with balanced functional group chemistry. Likewise, a set of 20,000 MOFs was found to have maximally balanced metal chemistry. Incorporating more than 20,000 MOFs in this set started to result in a decrease in balance. These “diverse subsets” are provided in the Supporting Information and labelled the ARC-MOF diverse-functional-group and diverse-metal subsets. Since the ligand chemistry and geometry are well balanced throughout ARC-MOF, balanced sets are not provided for these features. We anticipate that the ARC-MOF diverse subsets will be

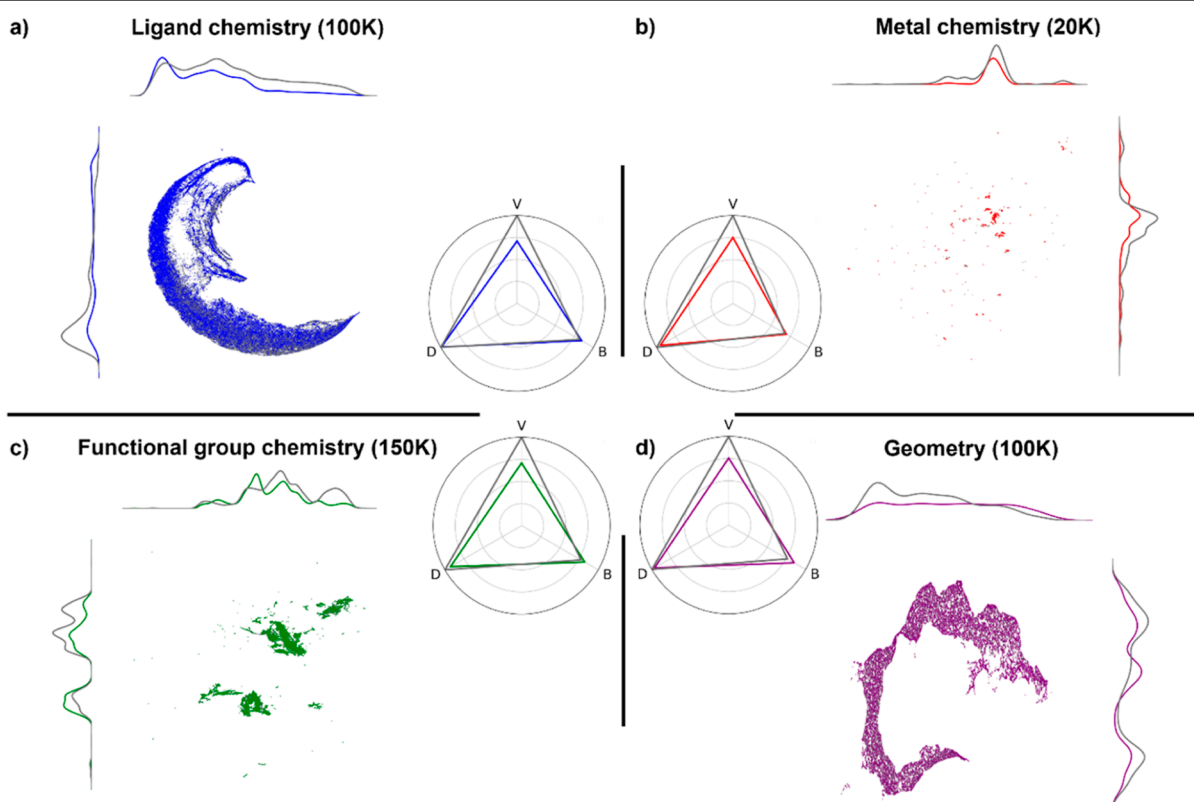


Figure 4. Two-dimensional UMAP projection of descriptors of a varying number of MOFs sampled using farthest point sampling and corresponding radar plots showing diversity metrics for the (a) ligand chemistry; (b) metal chemistry; (c) functional group chemistry; and (d) geometry. RAC descriptors were used for the chemical diversity (a–c), and geometric descriptors were used for the geometry (d). Structures present in ARC-MOF (~280 K MOFs) are represented by colored points, which is overlaid on the entire design space (~480 K MOFs), represented by gray points. The diversity metrics shown on the radar plots are variety (V), disparity (D), and balance (B), where gray again represents the entire set of MOFs and color represents the ARC-MOF subset. Only functionalized MOFs were considered for (c), and only MOFs with non-zero accessible surface area are shown in (d).

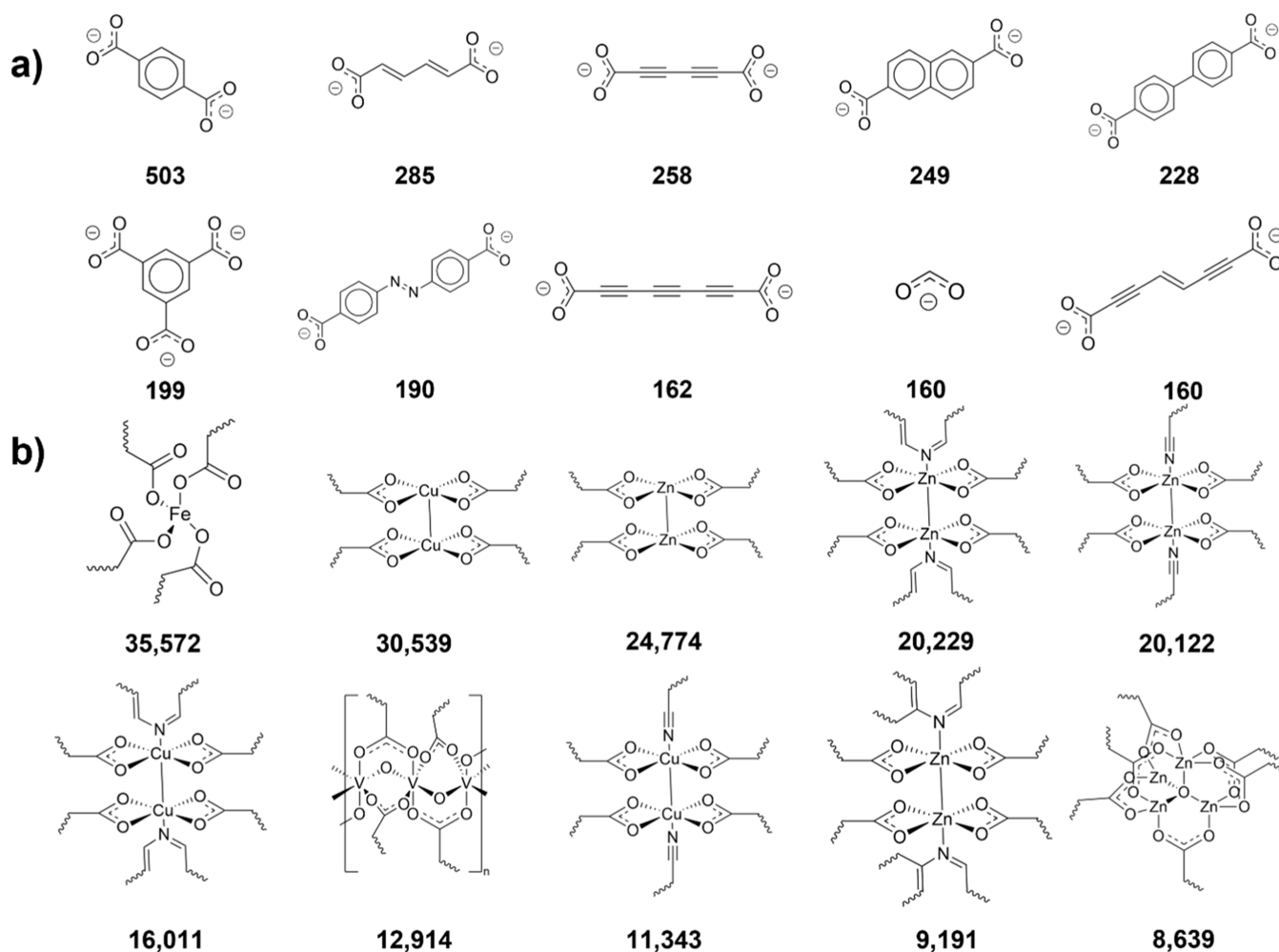


Figure 5. Top 10 most common (a) organic and (b) inorganic substructures in ARC-MOF, as determined through a clustering analysis of the (a) ligand and (b) metal-center RAC descriptors of the entire database. The numbers under each substructure represent the frequency of the respective substructure in ARC-MOF.

useful for training ML models which rely particularly on these types of chemistry, (e.g., applications in catalysis or low-pressure gas adsorption). It is important to note that while the diverse-metal subset is well balanced and varied with respect to metal chemistry, it is not necessarily expected to be well varied with respect to the ligand chemistry and geometry, for example.

Figure 4 shows how in particular for metal chemistry, a small fraction of MOFs (~5%) can sufficiently represent the diversity of the entire space and demonstrates the high imbalance of metal chemistry in the available (hypothetical) databases. The same conclusion can be drawn from Figure 5, which shows the 10 most common (a) organic and (b) inorganic substructures present in ARC-MOF and their corresponding frequency of occurrence in ARC-MOF. The top 10 most common organic substructures comprise ~1% of the entire ARC-MOF database, while the top 10 most common inorganic substructures are found in ~67% of the entire ARC-MOF database. Furthermore, the top 10 inorganic substructures are chemically similar, being composed primarily of variations of copper and zinc paddlewheels. A similar result is observed for the entire design space, in which 60% of the MOFs contain the top 10 most common inorganic substructures (Figure S4). If the same analysis is carried out on

the diverse-metal subset (20 K MOFs), only ~2% of the MOFs contain the top 10 most common inorganic substructures. Consequently, we conclude that it is critical that future ML and screening studies of MOFs curate their own data sets from MOF databases in a way that maximizes not only the number of structures in the data set but also the balance of the descriptor space of interest (e.g., the metal chemistry). Otherwise, it is unlikely such models will extend to these MOFs with underrepresented descriptor vectors, if these descriptors correlate strongly with the target property of interest.

REPEAT Charge Analysis. Partial atomic charges were derived from a DFT calculation of each MOF using the REPEAT method of Campaña et al.,³⁸ which is an ESP-fitted charge method. In these methods, the ESP from a first-principles (DFT) calculation is evaluated on grid points outside atomic radii of the atoms in the system. The partial charges on the atoms are then fit to minimize the difference between the ESP due to the DFT calculation and that due to the point charges themselves at each of the grid points. Histograms showing the distribution of the REPEAT charges in ARC-MOF for the most common elements are shown in Figure 6. REPEAT charge statistics for all elements present in the database are available in Table S5.

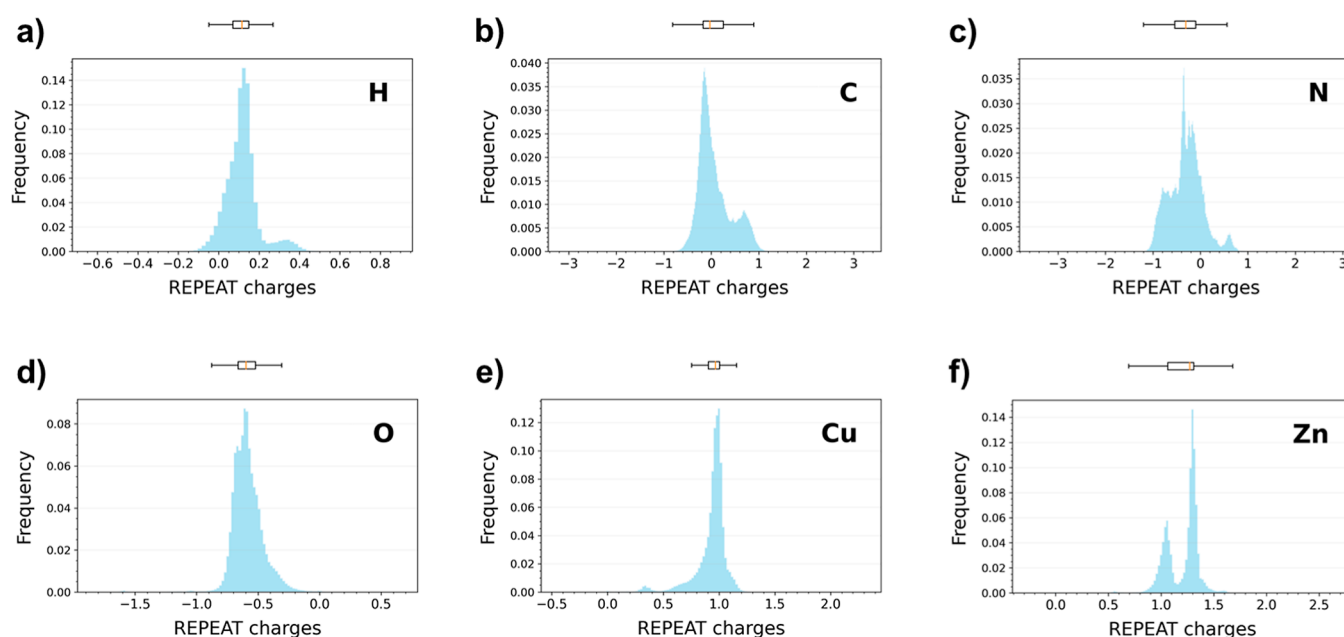


Figure 6. Histograms presenting the distribution of calculated REPEAT charges by element for (a) H, (b) C, (c) N, (d) O, (e) Cu, and (f) Zn. Each histogram represents the cumulative total of atoms extracted from all MOFs in the ARC–MOF database. The charges are given in units of e . Overhead boxplots illustrate the position of the REPEAT charge median (yellow line) and the interquartile range (box width) for each element.

To evaluate the importance of DFT-quality charges for atomistic simulations, we obtained gas uptake data for CO_2 and H_2S under various conditions using GCMC simulations with two different charge-assignment methods: QEq and REPEAT. CO_2 pressures of 0.15 and 1 bar were chosen due to their relevance to post-combustion carbon capture conditions¹⁰¹ and 0.4 mbar due to its relevance to direct-air carbon capture conditions.¹⁰² To demonstrate the importance of the charge-assignment method on gas adsorption simulations of highly polar guests, we performed a similar analysis on H_2S . Pressures ranging from 1 to 0.001 bar (1000 ppm) were chosen as a 1000 ppm concentration of H_2S is both the immediate lethal concentration in humans¹⁰³ and is frequently observed in biogas applications.¹⁰⁴

In this analysis, we treat REPEAT as being the most reliable charge method for obtaining gas uptakes using GCMC simulations. Therefore, linear regressions were performed between gas uptakes obtained using REPEAT charges and gas uptakes obtained using QEq. These plots and additional statistical information are available in the [Supporting Information](#). Slope, R^2 correlation, and Spearman rank coefficients from these plots are given in [Table 5](#). In this case, slope is considered a measure of accuracy while R^2 correlation is considered a measure of precision. Both metrics are equal unity when the gas uptakes obtained using the empirical charge-assignment method (i.e., QEq) are equal to the gas uptakes obtained using the REPEAT partial charges. When the rank coefficient is equal to unity, MOFs ordered by uptake derived from REPEAT is conserved in QEq data.

[Table 5](#) demonstrates that charge dependence on the gas uptake increases greatly at low pressures and for highly polar guests. Particularly, QEq is only able to reasonably reproduce gas uptakes obtained using REPEAT charges when the pressure is greater than or equal to 1 bar ($m = 0.90$ for CO_2 and $m = 0.84$ for H_2S). When considering hydrogen sulfide at low pressures (0.001 bar), QEq performs poorly in reproducing the gas uptake as simulated using REPEAT

Table 5. Correlation Metrics between GCMC Uptakes Simulated Using MEPO–QEq vs REPEAT Charges for 10,000 MOFs^a

guest	pressure (bar)	linear regression parameters		
		slope	R^2	Spearman R
CO_2	1.00	0.90	0.92	0.97
	0.15	0.86	0.94	0.97
	0.0004	0.22	0.22	0.96
H_2S	1.00	0.84	0.85	0.93
	0.1	0.63	0.75	0.93
	0.01	0.28	0.46	0.91
	0.001	0.07	0.03	0.90

^aThe R^2 correlation parameter is determined from the Pearson R correlation coefficient. The Spearman R parameter is determined from the Spearman rank correlation coefficient.

charges ($m = 0.07$ and $R^2 = 0.03$). This result underpins the value of a database with DFT-derived partial atomic charges for GCMC simulations where electrostatic interactions dominate such as simulations performed at low pressure, or for simulations involving polar guests such as water. This result is expected since at high pressure, the MOF becomes saturated with gas, so the MOF geometry is the most important factor affecting the gas uptake. In the cases of low chemical potential of gas (low pressure), interactions between the framework and guest molecules (and therefore, the quality partial atomic charges) become increasingly important for determining the gas uptake. Heatmaps of GCMC uptakes simulated using MEPO–QEq versus REPEAT charges are available in the [Supporting Information](#) ([Figures S6 and S7](#)).

Machine Learning Case Study—Comparison of Training Sets of Varying Diversity. While it is reasonable to hypothesize that a balanced training set seems it should perform better than a ML model trained on biased data, it has yet to be demonstrated. There are many contributing factors to model performance beyond diversity, such as training set size

and whether bias in a particular part of the data even correlates with the target property. To evaluate the potential utility of a diverse training set, a ML model trained on the diverse-metal subset (MC model) was compared to a ML model trained on a random sample of ARC–MOF of similar size (ARC model). It should be once again emphasized that the diverse-metal subset is balanced in all aspects evaluated in this paper, by virtue of the ligand chemistry and geometry being balanced throughout all of ARC–MOF. However, for the same reason, the diverse-metal subset is not necessarily sufficiently varied and disparate with respect to ligand chemistry and geometry (in comparison to the entire ARC–MOF set). Thus, in this case study, metal partial charges were chosen as the target property as these should have greater dependence on the balance, variety, and disparity of metal chemistry of the data set compared to the diversity of the ligand chemistry and pore geometry. Ideally, one would prepare a data set which is perfectly balanced, varied, and disparate with respect to all types of chemistry and pore geometry, but this is challenging given the few examples of most metal SBUs in available MOF databases, compared to the overwhelming diversity of organic SBUs (as suggested by the occurrence of common substructures in Figure 5). Thus, in future, emphasis should be placed on generating MOFs with a wider variety of metal SBUs using diverse repositories like the HEALD SBU library.¹⁰⁵

In this case study, GBDT regressors were trained using the SOAP descriptor of each metal atom to predict the partial atomic charge of metal atoms in MOFs. Notably, each model gives only a single output which does not depend explicitly on atomic identity of the metal. To evaluate the performance of these ML models, a test set of MOFs from the diverse-metal set which are not in ARC–MOF (~7000 MOFs) was used. Since these models are intended only for comparison purposes between training sets, not much consideration was placed on the curation of the test set, other than requiring a variety of metals for analysis of prediction quality. Notably, since the MOFs come from the diverse-metal set, there should be minimal overlap with the MC model training set and ARC model training set.

Interestingly, a MAE of 0.14 e was observed for metal charge prediction for the MC model, compared to a MAE of 0.15 e for the ARC model. Such a marginal improvement seems to suggest a lack of importance of a balanced training set for this application. However, since the test set is not necessarily balanced with respect to the target property (i.e., charges), it is important to inspect the charge prediction by element. The seven most abundant metals in the test set are Zn, Cu, Ni, Co, Fe, Li, and Cd, which comprise 91% of the entire test set. Thus, the overall MAE is dominated by the prediction performance on these metals, which is shown in Table 6.

Table 6 demonstrates a nearly consistent marginal improvement in the prediction performance between the two models for these metals. However, there is still the question of how these models may differ in the prediction performance for more exotic metals which may possess a less consistent charge distribution. The cases where the MAE differs by >0.1 e between the MC model and ARC model are summarized in Table 7.

Table 7 shows a significant improvement in charge prediction, particularly for the more uncommon metals in the test set. Where there is a large disparity in charge prediction (>0.1 e), the MC model performs better, with the sole exception of Pb. The MC model is clearly more

Table 6. Comparison of MAE (e) of Predicted Charges between MC Model and ARC Model for the Top Seven Most Common Metals in the Test Set^a

metal	MC model	ARC model
Zn (9647)	0.06	0.07
Cu (5597)	0.13	0.15
Ni (3668)	0.16	0.14
Co (3537)	0.09	0.09
Fe (3095)	0.20	0.22
Li (2596)	0.09	0.16
Cd (2190)	0.09	0.14

^aThe number in parentheses for each metal shows the number of examples of each metal in the test set.

Table 7. Comparison of MAE (e) of Predicted Charges for Elements with a Difference in MAE of Greater Than 0.1 e and Considering Only Elements for Which There are At Least 10 Examples in the Test Set^a

metal	MC model	ARC model
La (98)	0.09	0.35
Ce (91)	0.08	0.36
Nd (85)	0.34	0.63
Eu (72)	0.21	0.50
Er (68)	0.26	0.48
Dy (50)	0.23	0.52
Pb (33)	0.59	0.37
Tb (26)	0.24	0.51
Pr (26)	0.23	0.58
Gd (23)	0.20	0.66
Ca (22)	0.07	0.20
Ho (12)	0.23	0.52
Mo (10)	0.24	0.90

^aThe number in parentheses for each metal shows the number of examples of each metal in the test set.

generalizable, demonstrating the utility of having a diverse training set. Future studies should inspect the effect of training set diversity on the model performance and generalizability, particularly when training data are abundant but not necessarily diverse.

Descriptors and Target Data. A final goal of this work was to compute descriptors and target data relevant to gas adsorption of MOFs to enable “out of the box” use of ARC–MOF in future screening or ML studies. To this end, we have computed AP-RDF descriptors for the entire database and RAC descriptors for most MOFs in ARC–MOF. RAC descriptors could not be computed for all MOFs in ARC–MOF due to limitations of the structural graph generation method implemented in MolSimplify. AP-RDFs have previously been shown to have good performance for gas adsorption studies of MOFs, particularly for CO₂ separations.^{47,48} In addition to these descriptors, target data were computed corresponding to gas adsorption properties (e.g., gas uptake, working capacity, and selectivity) of all MOFs in ARC–MOF for five gas separation processes, namely (a) methane purification, (b) post-combustion VSA, (c) pre-combustion PSA, (d) methane storage PSA, and (e) landfill gas VPSA. The specific conditions of these separations can be found in the Supporting Information. Geometric descriptors for each MOF have also been made available in the Supporting Information.

CONCLUSIONS

The ARC–MOF database contains ~280,000 thoroughly structure-checked MOFs which have been either experimentally characterized or computationally generated, spanning all publicly available MOF databases, with DFT-derived partial atomic charges for each MOF. Additionally, ARC–MOF contains pre-computed descriptors for out-of-the-box ML applications. An in-depth analysis of the diversity of ARC–MOF with respect to the currently mapped design space of MOFs was performed—a critical, yet commonly overlooked aspect in past publications of MOF databases. Chemical insights about ARC–MOF were drawn from this analysis—primarily that ARC–MOF sufficiently spans the overall chemical space, and that it is sufficiently balanced with respect to geometric properties, as well as ligand chemistry. However, ARC–MOF suffers from being highly unbalanced with respect to metal chemistry, a well-known flaw of current hMOF databases. Thus, a subset of the overall design space which is balanced with respect to metal chemistry has been curated, for ML or screening applications where metal chemistry may be important (e.g., chemisorptive processes). As a result of the high imbalance of metal chemistry in the hypothetical databases, we conclude that it is critical that future ML and screening studies of MOFs curate their own data sets from MOF databases in a way that maximizes not only the number of structures in the data set but also the balance of the descriptor space of interest (e.g., the metal chemistry). It was shown that farthest point sampling is an effective method to do this, and an efficient code to perform this sampling is provided in the [Supporting Information](#). A key goal of this work is not only for researchers to make use of ARC–MOF but also adapt the proposed methodology for curating balanced data sets for ML and screening applications. It was demonstrated that having a balanced data set can be useful for improving ML model generalizability, particularly when bias in the data is known to correlate with the target property. For example, a bias in metal chemistry of a MOF database may not affect ML model generalizability for high-pressure gas uptake prediction (where the MOF pore geometry is the most important characteristic for the gas uptake, and metal chemistry has little importance). In the same vein, we have shown the utility of the ARC–MOF for applications where high-quality partial atomic charges are desired (e.g., simulations involving polar guests or simulations at low pressure). In these types of simulations, there is very poor correlation between GCMC-simulated uptakes obtained using empirical charge-assignment methods compared to GCMC-simulated uptakes obtained using DFT-quality partial atomic charges.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.chemmater.2c02485>.

ARC–MOF database (CIF files with charges) and associated descriptors/adsorption target properties, predicted topologies, all structures used in the diversity analysis, clusters used in the diversity analysis, and balanced subsets available on Zenodo (<https://doi.org/10.5281/zenodo.6908727>) and the codes used for the diversity analysis, REPEAT calculations, GCMC simulations, AP-RDF/RAC descriptor calculations, and oxidation state assignment available on the uOttawa

Woo Lab GitHub page (<https://github.com/uwooolab>) (PDF)

AUTHOR INFORMATION

Corresponding Author

Tom K. Woo – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada; orcid.org/0000-0003-0073-3901; Email: Tom.Woo@uottawa.ca

Authors

Jake Burner – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada; orcid.org/0000-0002-2834-5866

Jun Luo – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Andrew White – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Adam Mirmiran – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Ohmin Kwon – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Peter G. Boyd – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Stephen Maley – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Marco Gibaldi – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada; orcid.org/0000-0002-5734-2250

Scott Simrod – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Victoria Ogden – Department of Chemistry and Biomolecular Sciences, University of Ottawa, Ottawa K1N 6N5, Canada

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.chemmater.2c02485>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

Financial support from the Natural Sciences and Engineering Research Council of Canada, the University of Ottawa, MITACS, and TotalEnergies is greatly appreciated, as well as the computing resources provided by the Canada Foundation for Innovation, TotalEnergies, Compute Canada, and the US DOE HPC4EI program.

REFERENCES

- (1) Farha, O. K.; Eryazici, I.; Jeong, N. C.; Hauser, B. G.; Wilmer, C. E.; Sarjeant, A.; Snurr, R. Q.; Nguyen, S. T.; Yazaydin, Ö.; Hupp, J. T. Metal–Organic Framework Materials with Ultrahigh Surface Areas: Is the Sky the Limit? *J. Am. Chem. Soc.* **2012**, *134*, 15016–15021.
- (2) O’Keeffe, M.; Peskov, M. A.; Ramsden, S. J.; Yaghi, O. M. The Reticular Chemistry Structure Resource (RCSR) Database of, and Symbols for, Crystal Nets. *Acc. Chem. Res.* **2008**, *41*, 1782–1789.
- (3) Yang, D.; Gates, B. C. Catalysis by Metal Organic Frameworks: Perspective and Suggestions for Future Research. *ACS Catal.* **2019**, *9*, 1779–1798.
- (4) Park, J. G.; Collins, B. A.; Darago, L. E.; Runčevski, T.; Ziebel, M. E.; Aubrey, M. L.; Jiang, H. Z. H.; Velasquez, E.; Green, M. A.; Goodpaster, J. D.; Long, J. R. Magnetic Ordering through Itinerant Ferromagnetism in a Metal–Organic Framework. *Nat. Chem.* **2021**, *13*, 594–598.

- (5) Jian, M.; Qiu, R.; Xia, Y.; Lu, J.; Chen, Y.; Gu, Q.; Liu, R.; Hu, C.; Qu, J.; Wang, H.; Zhang, X. Ultrathin Water-Stable Metal-Organic Framework Membranes for Ion Separation. *Sci. Adv.* **2020**, *6*, No. eaay3998.
- (6) Mokri, N.; Sepehri, Z.; Faninam, F.; Khaleghi, S.; Kazemi, N. M.; Hashemi, M. Chitosan-Coated Zn-Metal-Organic Framework Nanocomposites for Effective Targeted Delivery of LNA-Antisense MiR-224 to Colon Tumor: In Vitro Studies. *Gene Ther.* **2021**, *29*, 680–690.
- (7) Sadiq, M. M.; Batten, M. P.; Mulet, X.; Freeman, C.; Konstas, K.; Mardel, J. I.; Tanner, J.; Ng, D.; Wang, X.; Howard, S.; Hill, M. R.; Thornton, A. W. A Pilot-Scale Demonstration of Mobile Direct Air Capture Using Metal–Organic Frameworks. *Adv. Sustainable Syst.* **2020**, *4*, 2000101.
- (8) Wen, H. M.; Liao, C.; Li, L.; Alsalmeh, A.; Allothman, Z.; Krishna, R.; Wu, H.; Zhou, W.; Hu, J.; Chen, B. A Metal-Organic Framework with Suitable Pore Size and Dual Functionalities for Highly Efficient Post-Combustion CO₂ Capture. *J. Mater. Chem. A* **2019**, *7*, 3128–3134.
- (9) Burns, T. D.; Pai, K. N.; Subraveti, S. G.; Collins, S. P.; Krykunov, M.; Rajendran, A.; Woo, T. K. Prediction of MOF Performance in Vacuum Swing Adsorption Systems for Postcombustion CO₂ Capture Based on Integrated Molecular Simulations, Process Optimizations, and Machine Learning Models. *Environ. Sci. Technol.* **2020**, *54*, 4536–4544.
- (10) Schoedel, A.; Ji, Z.; Yaghi, O. M. The Role of Metal–Organic Frameworks in a Carbon-Neutral Energy Cycle. *Nat. Energy* **2016**, *1*, 16034.
- (11) Bae, Y.-S. S.; Snurr, R. Q. Development and Evaluation of Porous Materials for Carbon Dioxide Separation and Capture. *Angew. Chem., Int. Ed.* **2011**, *50*, 11586–11596.
- (12) McCaffrey, R.; Edwards, P.; Perilli, D.; Winskell, J. *Global Cement Magazine: Reaching New Heights*; Global Cement Magazine: Epsom, Surrey, U.K., 2020; pp 20–22.
- (13) Lin, J.-B.; Nguyen, T. T. T.; Vaidhyanathan, R.; Burner, J.; Taylor, J. M.; Durekova, H.; Akhtar, F.; Mah, R. K.; Ghaffari-Nik, O.; Marx, S.; Fylstra, N.; Iremonger, S. S.; Dawson, K. W.; Sarkar, P.; Hovington, P.; Rajendran, A.; Woo, T. K.; Shimizu, G. K. H. A Scalable Metal-Organic Framework as a Durable Physiosorbent for Carbon Dioxide Capture. *Science* **2021**, *374*, 1464–1469.
- (14) Meng, Y. S.; Arroyo-de Dompablo, M. E. First Principles Computational Materials Design for Energy Storage Materials in Lithium Ion Batteries. *Energy Environ. Sci.* **2009**, *2*, 589–609.
- (15) Jain, A.; Shin, Y.; Persson, K. A. Computational Predictions of Energy Materials Using Density Functional Theory. *Nat. Rev. Mater.* **2016**, *1*, 15004.
- (16) Düren, T.; Bae, Y. S.; Snurr, R. Q. Using Molecular Simulation to Characterise Metal-Organic Frameworks for Adsorption Applications. *Chem. Soc. Rev.* **2009**, *38*, 1237–1247.
- (17) Jennings, P. C.; Lysgaard, S.; Hummelshøj, J. S.; Vegge, T.; Bligaard, T. Genetic Algorithms for Computational Materials Discovery Accelerated by Machine Learning. *npj Comput. Mater.* **2019**, *5*, 46.
- (18) Li, J.; Liu, J.; Baronett, S. A.; Liu, M.; Wang, L.; Li, R.; Chen, Y.; Li, D.; Zhu, Q.; Chen, X. Q. Computation and Data Driven Discovery of Topological Phononic Materials. *Nat. Commun.* **2021**, *12*, 1204.
- (19) Jablonka, K. M.; Ongari, D.; Moosavi, S. M.; Smit, B. Big-Data Science in Porous Materials: Materials Genomics and Machine Learning. *Chem. Rev.* **2020**, *120*, 8066–8129.
- (20) Boyd, P. G.; Chidambaram, A.; García-Díez, E.; Ireland, C. P.; Daff, T. D.; Bounds, R.; Gladysiak, A.; Schouwink, P.; Moosavi, S. M.; Maroto-Valer, M. M.; Reimer, J. A.; Navarro, J. A. R.; Woo, T. K.; Garcia, S.; Stylianou, K. C.; Smit, B. Data-Driven Design of Metal–Organic Frameworks for Wet Flue Gas CO₂ Capture. *Nature* **2019**, *576*, 253–256.
- (21) Moghadam, P. Z.; Li, A.; Wiggin, S. B.; Tao, A.; Maloney, A. G. P.; Wood, P. A.; Ward, S. C.; Fairen-Jimenez, D. Development of a Cambridge Structural Database Subset: A Collection of Metal–Organic Frameworks for Past, Present, and Future. *Chem. Mater.* **2017**, *29*, 2618–2625.
- (22) Groom, C. R.; Bruno, I. J.; Lightfoot, M. P.; Ward, S. C. The Cambridge Structural Database. *Acta Crystallogr., Sect. B: Struct. Sci., Cryst. Eng. Mater.* **2016**, *72*, 171–179.
- (23) Li, A.; Perez, R. B.; Wiggin, S.; Ward, S. C.; Wood, P. A.; Fairen-Jimenez, D. The Launch of a Freely Accessible MOF CIF Collection from the CSD. *Matter* **2021**, *4*, 1105–1106.
- (24) Chung, Y. G.; Camp, J.; Haranczyk, M.; Sikora, B. J.; Bury, W.; Krungleviciute, V.; Yildirim, T.; Farha, O. K.; Sholl, D. S.; Snurr, R. Q. Computation-Ready, Experimental Metal–Organic Frameworks: A Tool To Enable High-Throughput Screening of Nanoporous Crystals. *Chem. Mater.* **2014**, *26*, 6185–6192.
- (25) Nazarian, D.; Camp, J. S.; Sholl, D. S. A Comprehensive Set of High-Quality Point Charges for Simulations of Metal-Organic Frameworks. *Chem. Mater.* **2016**, *28*, 785.
- (26) Nazarian, D.; Camp, J. S.; Chung, Y. G.; Snurr, R. Q.; Sholl, D. S. Large-Scale Refinement of Metal-Organic Framework Structures Using Density Functional Theory. *Chem. Mater.* **2017**, *29*, 2521–2528.
- (27) Chung, Y. G.; Haldoupis, E.; Bucior, B. J.; Haranczyk, M.; Lee, S.; Zhang, H.; Vogiatzis, K. D.; Milisavljevic, M.; Ling, S.; Camp, J. S.; Slater, B.; Siepmann, J. I.; Sholl, D. S.; Snurr, R. Q. Advances, Updates, and Analytics for the Computation-Ready, Experimental Metal–Organic Framework Database: CoRE MOF 2019. *J. Chem. Eng. Data* **2019**, *64*, 5985–5998.
- (28) Chen, T.; Manz, T. A. Identifying Misbonded Atoms in the 2019 CoRE Metal-Organic Framework Database. *RSC Adv.* **2020**, *10*, 26944–26951.
- (29) Wilmer, C. E.; Leaf, M.; Lee, C. Y.; Farha, O. K.; Hauser, B. G.; Hupp, J. T.; Snurr, R. Q. Large-Scale Screening of Hypothetical Metal–Organic Frameworks. *Nat. Chem.* **2012**, *4*, 83–89.
- (30) Aghaji, M. Z.; Fernandez, M.; Boyd, P. G.; Daff, T. D.; Woo, T. K. Quantitative Structure–Property Relationship Models for Recognizing Metal Organic Frameworks (MOFs) with High CO₂ Working Capacity and CO₂/CH₄ Selectivity for Methane Purification. *Eur. J. Inorg. Chem.* **2016**, *2016*, 4505–4511.
- (31) Colón, Y. J.; Gómez-Gualdrón, D. A.; Snurr, R. Q. Topologically Guided, Automated Construction of Metal–Organic Frameworks and Their Evaluation for Energy-Related Applications. *Cryst. Growth Des.* **2017**, *17*, 5801–5810.
- (32) Lan, Y.; Yan, T.; Tong, M.; Zhong, C. Large-Scale Computational Assembly of Ionic Liquid/MOF Composites: Synergistic Effect in the Wire-Tube Conformation for Efficient CO₂/CH₄ Separation. *J. Mater. Chem. A* **2019**, *7*, 12556–12564.
- (33) Gurnani, R.; Yu, Z.; Kim, C.; Sholl, D. S.; Ramprasad, R. Interpretable Machine Learning-Based Predictions of Methane Uptake Isotherms in Metal-Organic Frameworks. *Chem. Mater.* **2021**, *33*, 3543.
- (34) Boyd, P. G.; Woo, T. K. A Generalized Method for Constructing Hypothetical Nanoporous Materials of Any Net Topology from Graph Theory. *CrystEngComm* **2016**, *18*, 3777–3792.
- (35) Collins, S. P.; Daff, T. D.; Piotrowski, S. S.; Woo, T. K. Materials Design by Evolutionary Optimization of Functional Groups in Metal-Organic Frameworks. *Sci. Adv.* **2016**, *2*, 1600954.
- (36) Majumdar, S.; Moosavi, S. M.; Jablonka, K. M.; Ongari, D.; Smit, B. Diversifying Databases of Metal Organic Frameworks for High-Throughput Computational Screening. *ACS Appl. Mater. Interfaces* **2021**, *13*, 61004–61014.
- (37) Moosavi, S. M.; Nandy, A.; Jablonka, K. M.; Ongari, D.; Janet, J. P.; Boyd, P. G.; Lee, Y.; Smit, B.; Kulik, H. J. Understanding the Diversity of the Metal-Organic Framework Ecosystem. *Nat. Commun.* **2020**, *11*, 4068.
- (38) Campaña, C.; Mussard, B.; Woo, T. K. Electrostatic Potential Derived Atomic Charges for Periodic Systems Using a Modified Error Functional. *J. Chem. Theory Comput.* **2009**, *5*, 2866–2878.
- (39) Ongari, D.; Boyd, P. G.; Kadioglu, O.; Mace, A. K.; Keskin, S.; Smit, B. Evaluating Charge Equilibration Methods To Generate

Electrostatic Fields in Nanoporous Materials. *J. Chem. Theory Comput.* **2019**, *15*, 382.

(40) Rappe, A. K.; Goddard, W. A. Charge Equilibration for Molecular-Dynamics Simulations. *J. Phys. Chem.* **1991**, *95*, 3358–3363.

(41) Nistor, R. A.; Polihronov, J. G.; Müser, M. H.; Mosey, N. J. A Generalization of the Charge Equilibration Method for Nonmetallic Materials. *J. Chem. Phys.* **2006**, *125*, 094108.

(42) Wilmer, C. E.; Kim, K. C.; Snurr, R. Q. An Extended Charge Equilibration Method. *J. Phys. Chem. Lett.* **2012**, *3*, 2506–2511.

(43) Manz, T. A.; Sholl, D. S. Chemically Meaningful Atomic Charges That Reproduce the Electrostatic Potential in Periodic and Nonperiodic Materials. *J. Chem. Theory Comput.* **2010**, *6*, 2455–2468.

(44) Kadantsev, E. S.; Boyd, P. G.; Daff, T. D.; Woo, T. K. Fast and Accurate Electrostatics in Metal Organic Frameworks with a Robust Charge Equilibration Parameterization for High-Throughput Virtual Screening of Gas Adsorption. *J. Phys. Chem. Lett.* **2013**, *4*, 3056–3061.

(45) Collins, S. P.; Woo, T. K. Split-Charge Equilibration Parameters for Generating Rapid Partial Atomic Charges in Metal–Organic Frameworks and Porous Polymer Networks for High-Throughput Screening. *J. Phys. Chem. C* **2017**, *121*, 903–910.

(46) Yao, Z.; Sánchez-Lengeling, B.; Bobbitt, N. S.; Bucior, B. J.; Kumar, S. G. H.; Collins, S. P.; Burns, T.; Woo, T. K.; Farha, O. K.; Snurr, R. Q.; Aspuru-Guzik, A. Inverse Design of Nanoporous Crystalline Reticular Materials with Deep Generative Models. *Nat. Mach. Intell.* **2021**, *3*, 76–86.

(47) Burner, J.; Schwiedrzik, L.; Krykunov, M.; Luo, J.; Boyd, P. G.; Woo, T. K. High-Performing Deep Learning Regression Models for Predicting Low-Pressure CO₂ Adsorption Properties of Metal–Organic Frameworks. *J. Phys. Chem. C* **2020**, *124*, 27996.

(48) Dureckova, H.; Krykunov, M.; Aghaji, M. Z.; Woo, T. K. Robust Machine Learning Models for Predicting High CO₂ Working Capacity and CO₂/H₂ Selectivity of Gas Adsorption in Metal Organic Frameworks for Precombustion Carbon Capture. *J. Phys. Chem. C* **2019**, *123*, 4133.

(49) Qiao, Z.; Peng, C.; Zhou, J.; Jiang, J. High-Throughput Computational Screening of 137953 Metal–Organic Frameworks for Membrane Separation of a CO₂/N₂/CH₄ Mixture. *J. Mater. Chem. A* **2016**, *4*, 15904–15912.

(50) Wang, R.; Zhong, Y.; Bi, L.; Yang, M.; Xu, D. Accelerating Discovery of Metal–Organic Frameworks for Methane Adsorption with Hierarchical Screening and Deep Learning. *ACS Appl. Mater. Interfaces* **2020**, *12*, 52797–52807.

(51) Rosen, A. S.; Iyer, S. M.; Ray, D.; Yao, Z.; Aspuru-Guzik, A.; Gagliardi, L.; Notestein, J. M.; Snurr, R. Q. Machine Learning the Quantum-Chemical Properties of Metal–Organic Frameworks for Accelerated Materials Discovery. *Matter* **2021**, *4*, 1578–1597.

(52) Rosen, A. S.; Fung, V.; Huck, P.; O'Donnell, C. T.; Horton, M. K.; Truhlar, D. G.; Persson, K. A.; Notestein, J. M.; Snurr, R. Q. High-Throughput Predictions of Metal–Organic Framework Electronic Properties: Theoretical Challenges, Graph Neural Networks, and Data Exploration. *npj Comput. Mater.* **2022**, *8*, 112.

(53) Raza, A.; Sturluson, A.; Simon, C. M.; Fern, X. Message Passing Neural Networks for Partial Charge Assignment to Metal–Organic Frameworks. *J. Phys. Chem. C* **2020**, *124*, 19070–19082.

(54) Kancharlappalli, S.; Gopalan, A.; Haranczyk, M.; Snurr, R. Q. Fast and Accurate Machine Learning Strategy for Calculating Partial Atomic Charges in Metal–Organic Frameworks. *J. Chem. Theory Comput.* **2021**, *17*, 3052–3064.

(55) Pardakhti, M.; Nanda, P.; Srivastava, R. Impact of Chemical Features on Methane Adsorption by Porous Materials at Varying Pressures. *J. Phys. Chem. C* **2020**, *124*, 4534–4544.

(56) Li, Z.; Bucior, B. J.; Chen, H.; Haranczyk, M.; Siepmann, J. I.; Snurr, R. Q. Machine Learning Using Host/Guest Energy Histograms to Predict Adsorption in Metal–Organic Frameworks: Application to Short Alkanes and Xe/Kr Mixtures. *J. Chem. Phys.* **2021**, *155*, 014701.

(57) Anderson, R.; Biong, A.; Gómez-Gualdrón, D. A. Adsorption Isotherm Predictions for Multiple Molecules in MOFs Using the

Same Deep Learning Model. *J. Chem. Theory Comput.* **2020**, *16*, 1271–1283.

(58) Choi, S.; Sholl, D. S.; Medford, A. J. Gaussian Approximation of Dispersion Potentials for Efficient Featurization and Machine-Learning Predictions of Metal–Organic Frameworks. *J. Chem. Phys.* **2022**, *156*, 214108.

(59) Altintas, C.; Altundal, O. F.; Keskin, S.; Yildirim, R. Machine Learning Meets with Metal Organic Frameworks for Gas Storage and Separation. *J. Chem. Inf. Model.* **2021**, *61*, 2131–2146.

(60) Sonpal, A.; Afzal, M. A. F.; An, Y.; Chandrasekaran, A.; Halls, M. D. Benchmarking Machine Learning Descriptors for Crystals. *ACS Symp. Ser.* **2022**, *1416*, 111–126.

(61) Fernandez, M.; Trefiak, N. R. N. R.; Woo, T. K. T. K. Atomic Property Weighted Radial Distribution Functions Descriptors of Metal–Organic Frameworks for the Prediction of Gas Uptake Capacity. *J. Phys. Chem. C* **2013**, *117*, 14095–14105.

(62) Kresse, G.; Hafner, J. Ab Initio Molecular Dynamics for Liquid Metals. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1993**, *47*, 558.

(63) Kresse, G.; Joubert, D. From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1999**, *59*, 1758.

(64) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, 3865–3868.

(65) Blöchl, P. E. Projector Augmented-Wave Method. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1994**, *50*, 17953–17979.

(66) Huang, Y. G.; Mu, B.; Schoenecker, P. M.; Carson, C. G.; Karra, J. R.; Cai, Y.; Walton, K. S. A Porous Flexible Homochiral SrSi₂ Array of Single-Stranded Helical Nanotubes Exhibiting Single-Crystal-to-Single-Crystal Oxidation Transformation. *Angew. Chem., Int. Ed.* **2011**, *50*, 436–440.

(67) Isayev, O.; Oses, C.; Toher, C.; Gossett, E.; Curtarolo, S.; Tropsha, A. Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals. *Nat. Commun.* **2017**, *8*, 1–12.

(68) Ong, S. P.; Richards, W. D.; Jain, A.; Hautier, G.; Kocher, M.; Cholia, S.; Gunter, D.; Chevrier, V. L.; Persson, K. A.; Ceder, G. Python Materials Genomics (Pymatgen): A Robust, Open-Source Python Library for Materials Analysis. *Comput. Mater. Sci.* **2013**, *68*, 314–319.

(69) Cordero, B.; Gómez, V.; Platero-Prats, A. E.; Revés, M.; Echeverría, J.; Cremades, E.; Barragán, F.; Alvarez, S. Covalent Radii Revisited. *J. Chem. Soc., Dalton Trans.* **2008**, *21*, 2832–2838.

(70) Zoubritsky, L.; Coudert, F.-X. CrystalNets.jl: Identification of Crystal Topologies. *SciPost Chem.* **2022**, *1*, 005.

(71) Willems, T. F.; Rycroft, C. H.; Kazi, M.; Meza, J. C.; Haranczyk, M. Algorithms and Tools for High-Throughput Geometry-Based Analysis of Crystalline Porous Materials. *Microporous Mesoporous Mater.* **2012**, *149*, 134–141.

(72) Ioannidis, E. I.; Gani, T. Z. H.; Kulik, H. J. MolSimplify: A Toolkit for Automating Discovery in Inorganic Chemistry. *J. Comput. Chem.* **2016**, *37*, 2106–2117.

(73) Cottrell, T. L.; Sutton, L. E. Covalency, Electronegativity and Electronegativity. *Proc. R. Soc. London, Ser. A* **1951**, *207*, 49–63.

(74) McInnes, L.; Healy, J.; Melville, J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. **2018**, arXiv:1802.03426. arXiv e-prints.

(75) Nolet, C. J.; Lafargue, V.; Raff, E.; Nanditale, T.; Oates, T.; Zedlewski, J.; Patterson, J. Bringing UMAP Closer to the Speed of Light with GPU Acceleration. **2020**, arXiv:2008.00325. arXiv e-prints.

(76) Campello, R. J. G. B.; Moulavi, D.; Zimek, A.; Sander, J. Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection. *ACM Trans. Knowl. Discov. Data* **2015**, *10*, 1–51.

(77) Wang, Y.; Yu, S.; Gu, Y.; Shun, J. Fast Parallel Algorithms for Euclidean Minimum Spanning Tree and Hierarchical Spatial Clustering. *Proceedings of the 2021 International Conference on Management of Data*; Association for Computing Machinery: New York, NY, USA, 2021; pp 1982–1995.

(78) Kennard, R. W.; Stone, L. A. Computer Aided Design of Experiments. *Technometrics* **1969**, *11*, 137–148.

- (79) García-Sánchez, A.; Ania, C. O.; Parra, J. B.; Dubbeldam, D.; Vlugt, T. J. H. H.; Krishna, R.; Calero, S. Transferable Force Field for Carbon Dioxide Adsorption in Zeolites. *J. Phys. Chem. C* **2009**, *113*, 8814–8820.
- (80) Kamath, G.; Lubna, N.; Potoff, J. J. Effect of Partial Charge Parametrization on the Fluid Phase Behavior of Hydrogen Sulfide. *J. Chem. Phys.* **2005**, *123*, 124505.
- (81) Provost, B. *An Improved N₂ Model for Predicting Gas Adsorption in MOFs and Using Molecular Simulation to Aid in the Interpretation of SSNMR Spectra of MOFs*; University of Ottawa, 2014.
- (82) Stubbs, J. M.; Potoff, J. J.; Siepmann, J. I. Transferable Potentials for Phase Equilibria. 6. United-Atom Description for Ethers, Glycols, Ketones, and Aldehydes. *J. Phys. Chem. B* **2004**, *108*, 17596–17605.
- (83) Belof, J. L.; Stern, A. C.; Space, B. An Accurate and Transferable Intermolecular Diatomic Hydrogen Potential for Condensed Phase Simulation. *J. Chem. Theory Comput.* **2008**, *4*, 1332–1337.
- (84) Prokhorenkova, L.; Gusev, G.; Vorobev, A.; Dorogush, A. V.; Gulin, A. CatBoost: Unbiased Boosting with Categorical Features. *Adv. Neural Inf. Process. Syst.* **2018**, *31*, 6638–6648.
- (85) Bartók, A. P.; Kondor, R.; Csányi, G. On Representing Chemical Environments. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2013**, *87*, 184115.
- (86) Himanen, L.; Jäger, M. O. J.; Morooka, E. V.; Federici Canova, F.; Ranawat, Y. S.; Gao, D. Z.; Rinke, P.; Foster, A. S. DScribe: Library of Descriptors for Machine Learning in Materials Science. *Comput. Phys. Commun.* **2020**, *247*, 106949.
- (87) De, S.; Bartók, A. P.; Csányi, G.; Ceriotti, M. Comparing Molecules and Solids across Structural and Alchemical Space. *Phys. Chem. Chem. Phys.* **2016**, *18*, 13754–13769.
- (88) Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; Vanderplas, J.; Passos, A.; Cournapeau, D.; Brucher, M.; Perrot, M.; Duchesnay, E. Scikit-Learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2012**, *12*, 2825–2830.
- (89) Raschka, S.; Patterson, J.; Nolet, C. Machine Learning in Python: Main Developments and Technology Trends in Data Science, Machine Learning, and Artificial Intelligence. *Information* **2020**, *11*, 193.
- (90) Rappe, A. K.; Casewit, C. J.; Colwell, K. S.; Goddard, W. A.; Skiff, W. M. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *J. Am. Chem. Soc.* **1992**, *114*, 10024–10035.
- (91) Lan, Y.; Han, X.; Tong, M.; Huang, H.; Yang, Q.; Liu, D.; Zhao, X.; Zhong, C. Materials Genomics Methods for High-Throughput Construction of COFs and Targeted Synthesis. *Nat. Commun.* **2018**, *9*, 1–10.
- (92) Anderson, R.; Rodgers, J.; Argueta, E.; Biong, A.; Gómez-Gualdrón, D. A. Role of Pore Chemistry and Topology in the CO₂ Capture Capabilities of MOFs: From Molecular Simulation to Machine Learning. *Chem. Mater.* **2018**, *30*, 6325–6337.
- (93) Gomez-Gualdrón, D. A.; Gutov, O. V.; Krungleviciute, V.; Borah, B.; Mondloch, J. E.; Hupp, J. T.; Yildirim, T.; Farha, O. K.; Snurr, R. Q. Computational Design of Metal–Organic Frameworks Based on Stable Zirconium Building Units for Storage and Delivery of Methane. *Chem. Mater.* **2014**, *26*, 5632–5639.
- (94) *Materials Studio*; Accelrys Software Inc.: San Diego, CA, 2009.
- (95) Chung, Y. G.; Gómez-Gualdrón, D. A.; Li, P.; Leperi, K. T.; Deria, P.; Zhang, H.; Vermeulen, N. A.; Stoddart, J. F.; You, F.; Hupp, J. T.; Farha, O. K.; Snurr, R. Q. In silico discovery of metal-organic frameworks for precombustion CO₂ capture using a genetic algorithm. *Sci. Adv.* **2016**, *2*, No. e1600909.
- (96) Li, S.; Chung, Y. G.; Simon, C. M.; Snurr, R. Q. High-Throughput Computational Screening of Multivariate Metal–Organic Frameworks (MTV-MOFs) for CO₂ Capture. *J. Phys. Chem. Lett.* **2017**, *8*, 6135–6141.
- (97) Anderson, G.; Schweitzer, B.; Anderson, R.; Gómez-Gualdrón, D. A. Attainable Volumetric Targets for Adsorption-Based Hydrogen Storage in Porous Crystals: Molecular Simulation and Machine Learning. *J. Phys. Chem. C* **2019**, *123*, 120–130.
- (98) Bao, Y.; Martin, R. L.; Haranczyk, M.; Deem, M. W. Silico Prediction of MOFs with High Deliverable Capacity or Internal Surface Area. *Phys. Chem. Chem. Phys.* **2015**, *17*, 11962–11973.
- (99) Lee, S.; Kim, B.; Cho, H.; Lee, H.; Lee, S. Y.; Cho, E. S.; Kim, J. Computational Screening of Trillions of Metal–Organic Frameworks for High-Performance Methane Storage. *ACS Appl. Mater. Interfaces* **2021**, *13*, 23647–23654.
- (100) Anderson, R.; Gómez-Gualdrón, D. A. Increasing Topological Diversity during Computational “Synthesis” of Porous Crystals: How and Why. *CrystEngComm* **2019**, *21*, 1653–1665.
- (101) Wang, M.; Lawal, A.; Stephenson, P.; Sidders, J.; Ramshaw, C. Post-Combustion CO₂ Capture with Chemical Absorption: A State-of-the-Art Review. *Chem. Eng. Res. Des.* **2011**, *89*, 1609–1624.
- (102) Petersen, H. A.; Luca, O. R. Application-Specific Thermodynamic Favorability Zones for Direct Air Capture of Carbon Dioxide. *Phys. Chem. Chem. Phys.* **2021**, *23*, 12533–12536.
- (103) Ng, P. C.; Hendry-Hofer, T. B.; Witeof, A. E.; Brenner, M.; Mahon, S. B.; Boss, G. R.; Haouzi, P.; Bebart, V. S. Hydrogen Sulfide Toxicity: Mechanism of Action, Clinical Presentation, and Countermeasure Development. *J. Med. Toxicol.* **2019**, *15*, 287–294.
- (104) Rasi, S.; Lätelä, J.; Rintala, J. Trace Compounds Affecting Biogas Energy Utilisation—A Review. *Energy Convers. Manage.* **2011**, *52*, 3369–3375.
- (105) Gibaldi, M.; Kwon, O.; White, A.; Burner, J.; Woo, T. K. The HEALED SBU Library of Chemically Realistic Building Blocks for Construction of Hypothetical Metal–Organic Frameworks. *ACS Appl. Mater. Interfaces* **2022**, *14*, 43372–43386.



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